LINMA2222: Stochastic optimal control and reinforcement learning

Part III: Stochastic systems

Guillaume Berger

Academic year 2026–2027

How to use these slides

These slides include many mathematical details. Read them in two passes:

1. **Start with the results.** Jump to the main statements and understand what they say, why they matter, and how to use them.
2. **Then revisit the assumptions and proofs.** Keep in mind that technical "qualification" conditions are needed to rule out edge or degenerate cases.

A useful first-pass model: In many cases, think of a **finite-state, irreducible, aperiodic Markov chain** with bounded costs, rather than memorizing every technical assumption.

This covers many regularity and ergodicity conditions; result-specific requirements (e.g., on step sizes or features) still need to be checked.

Theory and practice

Theoretical guarantees: A theorem states what is guaranteed under explicit assumptions. If an assumption fails, the conclusion may still hold, but the theorem no longer establishes it.

A framework for practice: Some assumptions are difficult or impossible to verify with the information available in practice. Theory often provides a framework for understanding and designing algorithms, rather than a precise recipe for applying them.

Empirical evidence: Algorithms can work well beyond these assumptions. Evaluate performance and stability across multiple runs and settings; successful experiments do not establish a general guarantee.

Practical modifications: Gradient clipping, feature normalization, and parameter projection can improve stability. Check whether the theoretical guarantees still apply to the modified algorithm.

Table of Contents

Stochastic systems and value equations

Least-squares minimization of TD residuals

TD(0) – learn from one transition

Eligibility traces and TD(λ) – propagating information backward

Off-policy learning – exploration and stability

Learning a control policy – policy iteration, SARSA and Q-learning

Learning a parametrized policy – policy gradients and actor–critic

☕ Going further – modern reinforcement learning algorithms

Table of Contents

Stochastic systems and value equations

Least-squares minimization of TD residuals

TD(0) – learn from one transition

Eligibility traces and TD(λ) – propagating information backward

Off-policy learning – exploration and stability

Learning a control policy – policy iteration, SARSA and Q-learning

Learning a parametrized policy – policy gradients and actor–critic

☕ Going further – modern reinforcement learning algorithms

Autonomous stochastic systems

System:

$$X(k+1) = F(X(k), W(k)), \qquad k \geq 0,$$

where $X(k) \in \mathcal{X}$ is the state and $\{W(k)\}_{k\geq 0} \subseteq \mathcal{W}$ is an i.i.d. noise sequence with individual law ν.

At each step, $W(k)$ is independent of the state history $\{X(j)\}_{j=0}^{k}$.

Consequently, for every initial state $X(0) = x$, $\{X(k)\}_{k\geq 0}$ is a time-homogeneous Markov process: the conditional distribution of $X(k+1)$ given the state history depends only on $X(k)$.

Transition operator for an autonomous system

For a measurable state function f, define

$$(Pf)(x) := \mathbb{E}[f(X(k+1)) \mid X(k) = x] = \int_{\mathcal{W}} f(F(x,w))\,\nu(dw).$$

Time homogeneity makes this operator independent of k. Use this notation whenever the expectation is well defined, in particular for nonnegative f or integrable $|f|$.

Iterating the transition operator gives, for every integer $T \geq 0$,

$$(P^T f)(x) = \mathbb{E}[f(X(T)) \mid X(0) = x], \qquad P^0 f = f.$$

For a measurable set A, write $P^T(x, A) = (P^T \mathbf{1}_A)(x)$ for the T-step transition probability.

Invariant distribution and ergodic law of large numbers

Invariant distribution: A probability measure π satisfying $\pi P = \pi$, meaning that, for every bounded measurable f,

$$\int_{\mathcal{X}} (Pf)(x)\,\pi(dx) = \int_{\mathcal{X}} f(x)\,\pi(dx).$$

Ergodic law of large numbers: For each $f \in L^1(\pi)$ and every initial state $X(0) = x$,

$$\frac{1}{T}\sum_{k=0}^{T-1} f(X(k)) \xrightarrow[T\to\infty]{\text{a.s.}} \int_{\mathcal{X}} f(y)\,\pi(dy).$$

Remark

A sufficient condition for ELLN is *positive Harris recurrence*: the chain has an invariant probability measure and visits every set of positive π-measure almost surely from every initial state.

Convergence to equilibrium – total variation and rates

Let π be invariant. Total variation measures the largest discrepancy over measurable sets:

$$\|\mu - \nu\|_{\mathrm{TV}} := \sup_{A \text{ measurable}} \big|\mu(A) - \nu(A)\big|.$$

Convergence in total variation:

$$\|P^k(x,\cdot) - \pi\|_{\mathrm{TV}} \longrightarrow 0 \quad \text{for every } x \in \mathcal{X}.$$

Geometric ergodicity: For some $r \in (0,1)$ and finite $M(x)$,

$$\|P^k(x,\cdot) - \pi\|_{\mathrm{TV}} \leq M(x) r^k \quad \text{for every } x \in \mathcal{X},\ k \geq 0.$$

Uniform geometric ergodicity: For some $M < \infty$ and $r \in (0,1)$,

$$\|P^k(x,\cdot) - \pi\|_{\mathrm{TV}} \leq M r^k \quad \text{for every } x \in \mathcal{X},\ k \geq 0.$$

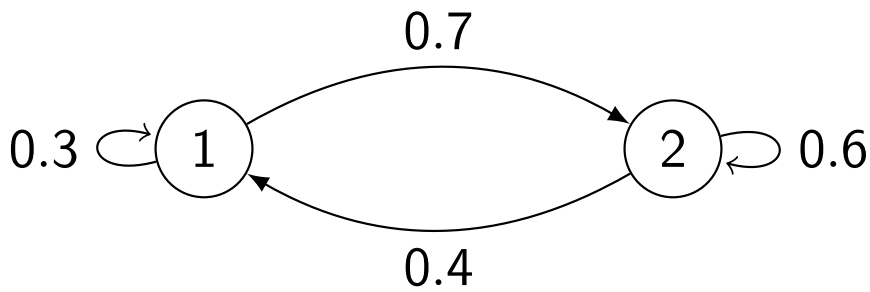
Two examples of autonomous stochastic systems

Linear system:

$$X(k+1) = FX(k) + W(k),$$

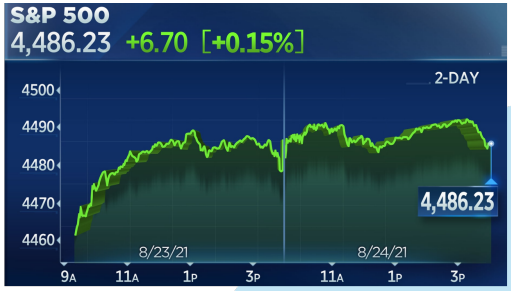
where $F \in \mathbb{R}^{n \times n}$ is stable, $\mathcal{X} = \mathbb{R}^n$, $W(k) \sim \mathcal{N}(0, \Sigma)$.

Markov chain:



where $\mathcal{X} = \{1, 2\}$.

Applications of stochastic systems

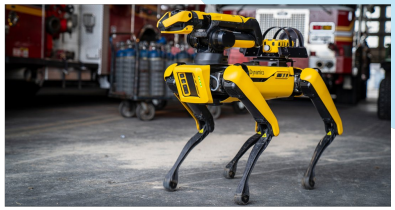


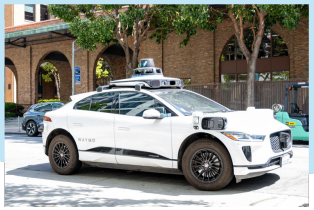




Stochasticity
in
Systems and Control









Discounted cost and value function

Consider a measurable cost function $c : \mathcal{X} \to \mathbb{R}_{\geq 0}$ and a discount factor $0 < \gamma < 1$.

The value function gives the *expected cumulative discounted cost*.

Value function:

$$h(x) := \sum_{k=0}^{\infty} \gamma^k (P^k c)(x).$$

Average cost and relative value function

Assume an invariant state distribution π. Consider a measurable cost function $c : \mathcal{X} \to \mathbb{R}_{\geq 0}$.

Average cost:

$$\eta := \int_{\mathcal{X}} c(x)\, \pi(dx).$$

Relative value function:

$$h(x) := \lim_{T \to \infty} \sum_{k=0}^{T-1} (P^k(c - \eta))(x),$$

if the series converges for every $x \in \mathcal{X}$.

Long-run averaged cost and average cost

Theorem

Let π be an invariant probability measure, $c \in L^1(\pi)$, and $\eta = \int c \, d\pi$. Assume the ergodic law of large numbers holds. Then, for any initial state $X(0) = x$,

$$\frac{1}{T} \sum_{k=0}^{T-1} c(X(k)) \xrightarrow[T \to \infty]{\text{a.s.}} \eta.$$

The proof is direct from ergodic law of large numbers.

Bellman equation for discounted cost

Proposition

The value function h is the minimal nonnegative solution of the Bellman equation:

$$\boxed{v(x) = c(x) + \gamma(Pv)(x) \quad \text{for every } x \in \mathcal{X}.}$$

A solution v, with v real-valued and measurable, equals h if, for every $x \in \mathcal{X}$,

$$(P^T|v|)(x) < \infty \quad \forall T \geq 0,$$

and the following transversality condition holds:

$$\gamma^T(P^T v)(x) \xrightarrow[T\to\infty]{} 0.$$

Bellman equation: proof (1/2)

Define the finite-horizon discounted values

$$h_T(x) := \sum_{j=0}^{T-1} \gamma^j (P^j c)(x) \xrightarrow[T\to\infty]{} h(x).$$

Splitting off the first cost and using composition of P gives

$$h_{T+1}(x) = c(x) + \gamma \sum_{j=0}^{T-1} \gamma^j (P^{j+1} c)(x) = c(x) + \gamma (P h_T)(x).$$

Since $c \geq 0$, we have $h_T \uparrow h$. Monotone convergence yields

$$h(x) = c(x) + \gamma (P h)(x).$$

By time homogeneity, this is the claimed operator equation.

Bellman equation: proof (2/2)

Iterating the equation using the Markov and tower properties gives

$$v(x) = h_T(x) + \gamma^T (P^T v)(x).$$

For a nonnegative solution v, the final term is nonnegative, so

$$v(x) \geq h_T(x) \xrightarrow[T\to\infty]{} h(x).$$

Hence, h is the minimal nonnegative solution.

For a solution satisfying transversality, the final term in the first identity tends to zero, giving $v(x) = h(x)$.

Poisson equation for average cost

Assume finite average cost and relative value function. Assume the domination condition on the next slide.

Proposition

The pair (η, h) solves the Poisson equation:

$$\boxed{g + v(x) = c(x) + (Pv)(x) \quad \text{for every } x \in \mathcal{X}.}$$

A solution (g, v), with $g \in \mathbb{R}$ and v real-valued and measurable, satisfies $g = \eta$ and $v(x) = h(x) + L$, with $L \in \mathbb{R}$, if, for every $x \in \mathcal{X}$,

$$(P^T|v|)(x) < \infty \quad \forall T \geq 0.$$

and the following transversality condition holds:

$$(P^T v)(x) \xrightarrow[T\to\infty]{} L.$$

Poisson equation: proof (1/2)

Define the finite-horizon relative values

$$h_T(x) := \sum_{j=0}^{T-1} (P^j(c-\eta))(x) \xrightarrow[T\to\infty]{} h(x).$$

Sufficient domination condition: For some measurable $V \geq 0$,

$$|h_T(y)| \leq V(y) \quad \forall T, y, \qquad (PV)(x) < \infty \quad \forall x.$$

Splitting off the first cost and using the Markov property gives

$$h_{T+1}(x) = c(x) - \eta + (Ph_T)(x).$$

Dominated convergence now yields

$$\eta + h(x) = c(x) + (Ph)(x).$$

By time homogeneity, this is the claimed operator equation.

Poisson equation: proof (2/2)

Iterating $g + v = c + Pv$ gives

$$v(x) = h_T(x) + T(\eta - g) + (P^T v)(x).$$

Since $h_T(x) \to h(x)$ and $(P^T v)(x) \to L$, dividing by T yields $g = \eta$. Substituting back and letting $T \to \infty$ gives

$$v(x) = h(x) + L.$$

Why a normalization is needed: If v solves the equation, so does $v + a$ for any constant a. Unlike the discounted case, there is no factor γ^T forcing bounded terminal terms to vanish. More generally, differences w of solutions with the same scalar g satisfy $w(x) = (Pw)(x)$. Without further assumptions, such functions need not be constant.

Linear-quadratic example: Bellman and Poisson

Let $c(x) = x^\top Q x$. Define $h(x) = x^\top \Omega x + q$.

Bellman equation:

$$\begin{aligned}
&\Rightarrow\ x^\top \Omega x + q = x^\top Q x + \gamma \mathbb{E}_N[(Fx+N)^\top \Omega (Fx+N) + q] \\
&\Leftrightarrow\ x^\top \Omega x + q = x^\top Q x + \gamma x^\top F^\top \Omega F x + \gamma \operatorname{tr}(\Omega\Sigma) + \gamma q \\
&\Leftrightarrow\ \Omega = Q + \gamma F^\top \Omega F \quad \text{and} \quad q = \gamma \operatorname{tr}(\Omega\Sigma) + \gamma q.
\end{aligned}$$

Poisson equation:

$$x^\top \Omega x + q = x^\top Q x - \eta + x^\top F^\top \Omega F x + \operatorname{tr}(\Omega\Sigma) + q.$$

Hence,

$$\Omega = Q + F^\top \Omega F, \qquad \eta = \operatorname{tr}(\Omega\Sigma).$$

The constant q is arbitrary in the Poisson equation.

Two-state example: Bellman and Poisson

Let $c(1) = 0$ and $c(2) = 1$.

Bellman equation:

$$h(1) = 0 + \gamma 0.3h(1) + \gamma 0.7h(2)$$
$$h(2) = 1 + \gamma 0.4h(1) + \gamma 0.6h(2).$$

Thus $h(1) = \frac{0.7\gamma}{(1-\gamma)(1+0.1\gamma)}$ and $h(2) = \frac{1-0.3\gamma}{(1-\gamma)(1+0.1\gamma)}$.

Poisson equation:

$$h(1) = -\eta + 0.3h(1) + 0.7h(2),$$
$$h(2) = 1 - \eta + 0.4h(1) + 0.6h(2).$$

Thus $\eta = 7/11$ and $h(2) - h(1) = 10/11$.

Controlled stochastic systems

System:

$$X(k+1) = F(X(k), U(k), W(k)), \qquad k \geq 0,$$

where $X(k) \in \mathcal{X}$ is the state, $U(k) \in \mathcal{U}$ is the input, and $\{W(k)\}_{k \geq 0} \subseteq \mathcal{W}$ is an i.i.d. noise sequence with individual law ν.

At each step, $W(k)$ is independent of the state–action history $(X(0), U(0), \ldots, X(k), U(k))$.

Consequently, under a stationary policy ϕ (see next slide): for every initial state $X(0) = x$, the state process $\{X(k)\}_{k \geq 0}$ is a time-homogeneous Markov process; for every initial state–input pair $(X(0), U(0)) = (x, u)$, the joint process $\{(X(k), U(k))\}_{k \geq 0}$ is a time-homogeneous Markov process.

Stationary policies

Stationary policies use only the current state and the same rule at every time step. More precisely, it is represented by a measurable probability kernel $\phi(du \mid x)$ satisfying

$$\mathcal{L}(U(k) \mid X(0), U(0), \ldots, X(k)) = \phi(\cdot \mid X(k)).$$

If $\phi(\cdot \mid x)$ is supported on a singleton, the policy is called **deterministic**; otherwise it is called **randomized**.

Remark

Unless stated otherwise, all policies considered in these slides are assumed stationary.

Transition state operators for a controlled system

For a measurable state function f and a fixed action u, define

$$(P_u f)(x) := \mathbb{E}[f(X(k+1)) \mid X(k) = x, U(k) = u]$$
$$= \int_{\mathcal{W}} f(F(x, u, w))\, \nu(dw).$$

For a stationary policy ϕ, the transition operator of the closed-loop state process is

$$(P_\phi f)(x) = \int_{\mathcal{U}} (P_u f)(x)\, \phi(du \mid x).$$

All expectations are understood when well defined.

For a state–action function q, write $m_q(x) = \min_a q(x, a)$ whenever the minimum exists and is measurable. Thus $(P_u m_q)(x)$ is the expected minimum value of q at the next state under input u.

Transition joint operator for a controlled system

For a stationary policy ϕ and a measurable state–action function q, define

$$(\mathcal{P}_\phi q)(x,u) := P_u\left[y \mapsto \int_{\mathcal{U}} q(y,a)\,\phi(da \mid y)\right](x).$$

This is the transition operator of the closed-loop joint process.
Equivalently,

$$(\mathcal{P}_\phi q)(x,u) = \mathbb{E}^\phi[q(X(1),U(1)) \mid X(0)=x, U(0)=u].$$

Iterating gives, for every integer $T \geq 0$,

$$(\mathcal{P}_\phi^T q)(x,u) = \mathbb{E}^\phi[q(X(T),U(T)) \mid X(0)=x, U(0)=u],$$

with $\mathcal{P}_\phi^0 q = q$, whenever these expectations are well defined.

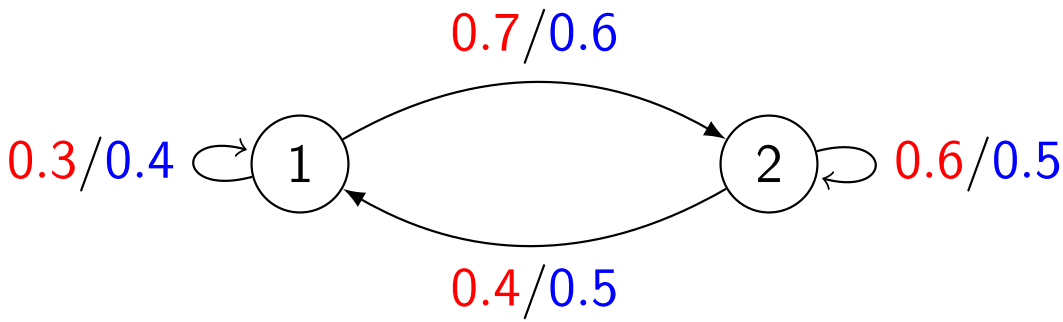
Two examples of controlled systems

Linear system:

$$X(k+1) = FX(k) + GU(k) + W(k),$$

where $F \in \mathbb{R}^{n\times n}$, $G \in \mathbb{R}^{n\times m}$, $\mathcal{X} = \mathbb{R}^n$, $\mathcal{U} = \mathbb{R}^m$, $W(k) \sim \mathcal{N}(0, \Sigma)$.

Markov decision process (MDP):



where $\mathcal{X} = \{1, 2\}$, $\mathcal{U} = \{\text{red}, \text{blue}\}$.

Discounted cost, value function, and Q-function

Consider a policy ϕ, a measurable cost function $c : \mathcal{X} \times \mathcal{U} \to \mathbb{R}_{\geq 0}$, and discount factor $0 < \gamma < 1$. Define the policy-averaged cost $c_\phi(x) = \int_{\mathcal{U}} c(x,u)\, \phi(du \mid x)$.

Value function:

$$h_\phi(x) := \sum_{k=0}^{\infty} \gamma^k (P_\phi^k c_\phi)(x).$$

Q-function:

$$Q_\phi(x,u) := \sum_{k=0}^{\infty} \gamma^k (\mathcal{P}_\phi^k c)(x,u).$$

Average cost and relative value function

Consider a policy ϕ and a measurable cost function $c : \mathcal{X} \times \mathcal{U} \to \mathbb{R}_{\geq 0}$, with policy-averaged cost c_ϕ defined as above. Assume closed-loop invariant state distribution π_ϕ.

Average cost:

$$\eta_\phi := \int_{\mathcal{X}} c_\phi(x)\, \pi_\phi(dx).$$

Relative value function:

$$h_\phi(x) := \lim_{T \to \infty} \sum_{k=0}^{T-1} (P_\phi^k (c_\phi - \eta_\phi))(x),$$

if the series converges for every $x \in \mathcal{X}$.

Average cost and relative Q-function

Relative Q-function:

$$Q_\phi(x,u) := \lim_{T\to\infty} \sum_{k=0}^{T-1} (\mathcal{P}_\phi^k(c-\eta_\phi))(x,u),$$

if the series converges for every $x \in \mathcal{X}$ and $u \in \mathcal{U}$.

Bellman Q-optimality – regularity assumptions

Let q be a real-valued measurable candidate. Assume:

Integrability: For every policy ϕ, initial pair $(x, u) \in \mathcal{X} \times \mathcal{U}$, and $T \geq 0$,

$$(\mathcal{P}_\phi^T |q|)(x, u) < \infty.$$

Transversality: For every policy ϕ and initial pair $(x, u) \in \mathcal{X} \times \mathcal{U}$,

$$\gamma^T (\mathcal{P}_\phi^T q)(x, u) \xrightarrow[T \to \infty]{} 0.$$

Attainment: A measurable minimizing policy exists:

$$\phi_\star(x) \in \arg\min_u q(x, u).$$

Bellman Q-optimality

Define $Q_\star(x,u) = \inf_\phi Q_\phi(x,u)$ for the optimal Q-function.

Proposition

Under the regularity conditions on the preceding slide, a real-valued measurable solution q of the Bellman optimality equation:

$$\boxed{q(x,u) = c(x,u) + \gamma(P_u m_q)(x) \quad \text{for all } x \in \mathcal{X} \text{ and } u \in \mathcal{U},}$$

satisfies

$$q = Q_{\phi_\star} = Q_\star.$$

Thus the minimizing policy $\phi_\star$ is optimal.

Bellman Q-optimality: proof

After the forced initial action, let $U(k)$ follow any policy ϕ. The minimum over next actions is no larger than its average under ϕ. Positivity of $\mathcal{P}_\phi$ and iteration of the optimality equation give

$$q(x,u) \le \sum_{k=0}^{T-1} \gamma^k (\mathcal{P}_\phi^k c)(x,u) + \gamma^T (\mathcal{P}_\phi^T q)(x,u).$$

The transversality condition and monotone convergence of the nonnegative cost sums give $q(x,u) \le Q_\phi(x,u)$. Every inequality is an equality for the minimizing policy $\phi_\star$, hence $q = Q_{\phi_\star} = Q_\star$.

Poisson Q-optimality – regularity assumptions

Let q be a real-valued measurable candidate. Assume:

Integrability: For every policy ϕ, initial pair $(x, u) \in \mathcal{X} \times \mathcal{U}$, and $T \geq 0$,

$$(\mathcal{P}_\phi^T |q|)(x, u) < \infty.$$

Transversality: For every policy ϕ and initial pair $(x, u) \in \mathcal{X} \times \mathcal{U}$,

$$\frac{1}{T}(\mathcal{P}_\phi^T q)(x, u) \xrightarrow[T \to \infty]{} 0.$$

Attainment: A measurable minimizing policy exists:

$$\phi_\star(x) \in \arg\min_u q(x, u).$$

Poisson Q-optimality

Assume finite average cost and relative value function for every policy ϕ. Define $\eta_\star = \inf_\phi \eta_\phi$ for the optimal average cost.

Proposition

Under the regularity conditions on the preceding slide, a real scalar g and a real-valued measurable function q, solution of the Poisson optimality equation:

$$\boxed{g + q(x,u) = c(x,u) + (P_u m_q)(x) \quad \text{for all } x \in \mathcal{X} \text{ and } u \in \mathcal{U},}$$

satisfy

$$g = \eta_{\phi_\star} = \eta_\star.$$

Thus the minimizing policy $\phi_\star$ is optimal for average cost.

Poisson Q-optimality: proof

After the forced initial action, let $U(k)$ follow any policy ϕ. The minimum over next actions is no larger than its average under ϕ. Positivity of $\mathcal{P}_\phi$ and iteration of the optimality equation give

$$Tg + q(x,u) \leq \sum_{k=0}^{T-1} (\mathcal{P}_\phi^k c)(x,u) + (\mathcal{P}_\phi^T q)(x,u).$$

Divide by T and use the transversality condition and convergence of the expected average cost to obtain $g \leq \eta_\phi$. Every inequality is an equality for the minimizing policy $\phi_\star$, hence $g = \eta_{\phi_\star} = \eta_\star$.

Relative Q-function: If additionally

$$(\mathcal{P}_{\phi_\star}^T q)(x,u) \xrightarrow[T\to\infty]{} L$$

for a common finite L, subtracting Tg in the equality gives $q = Q_{\phi_\star} + L$. Fixing the additive constant gives an optimal relative Q-function $Q_\star$.

Greedy policy improvement for discounted cost

Proposition

Given a policy ϕ, choose

$$\phi^+(x) \in \arg\min_u Q_\phi(x, u).$$

Then

$$h_{\phi^+}(x) \leq h_\phi(x), \qquad Q_{\phi^+}(x, u) \leq Q_\phi(x, u)$$

for every $(x, u) \in \mathcal{X} \times \mathcal{U}$.

Improvement is weak: the new policy may have the same value. If $h_\phi(x) = \min_u Q_\phi(x, u)$ everywhere, Q_ϕ solves the Q-optimality equation; under the regularity conditions, ϕ is optimal.

Policy improvement for discounted cost: proof

Conditioning on the first transition and using the greedy choice gives

$$c(x, \phi^+(x)) + \gamma(P_{\phi^+} h_\phi)(x) = \min_u Q_\phi(x, u) \le h_\phi(x).$$

Iterate this inequality along the new policy:

$$\sum_{k=0}^{T-1} \gamma^k (P_{\phi^+}^k c_{\phi^+})(x) + \gamma^T (P_{\phi^+}^T h_\phi)(x) \le h_\phi(x).$$

Drop the nonnegative terminal term and let $T \to \infty$ to obtain $h_{\phi^+} \le h_\phi$ by monotone convergence. For a forced first action,

$$\begin{aligned} Q_{\phi^+}(x, u) &= c(x, u) + \gamma(P_u h_{\phi^+})(x) \\ &\le c(x, u) + \gamma(P_u h_\phi)(x) = Q_\phi(x, u). \end{aligned}$$

Greedy policy improvement for average cost

Proposition

Given a policy ϕ, choose

$$\phi^+(x) \in \arg\min_u Q_\phi(x, u).$$

Then $\eta_{\phi^+} \leq \eta_\phi$. Improvement is strict if $\min_u Q_\phi(x, u) < h_\phi(x)$ on a positive π_{ϕ^+}-measure set.

This is an improvement in average cost, not necessarily in relative values. If $h_\phi(x) = \min_u Q_\phi(x, u)$ everywhere, (η_ϕ, Q_ϕ) solves the Q-optimality equation; under the regularity conditions, ϕ is average-optimal.

Policy improvement for average cost: proof

Conditioning on the first transition gives

$$Q_\phi(x,u) = c(x,u) - \eta_\phi + (P_u h_\phi)(x).$$

Define the nonnegative greedy improvement margin

$$d(x) := h_\phi(x) - \min_u Q_\phi(x,u) \geq 0.$$

For $u = \phi^+(x)$, rearrangement yields

$$c(x,\phi^+(x)) - \eta_\phi + (P_{\phi^+} h_\phi)(x) - h_\phi(x) = -d(x).$$

Average with respect to the invariant distribution π_{ϕ^+}. Invariance cancels the two relative-value terms, giving

$$\eta_{\phi^+} - \eta_\phi = -\pi_{\phi^+}(d) \leq 0.$$

The inequality is strict when $d > 0$ on a positive π_{ϕ^+}-measure set.

Table of Contents

Stochastic systems and value equations

Least-squares minimization of TD residuals

TD(0) – learn from one transition

Eligibility traces and TD(λ) – propagating information backward

Off-policy learning – exploration and stability

Learning a control policy – policy iteration, SARSA and Q-learning

Learning a parametrized policy – policy gradients and actor–critic

☕ Going further – modern reinforcement learning algorithms

Linear approximation and sampled TD residuals

Goal: Evaluate the value function of an autonomous system from observed transitions and cost values.

Idea: Write the observations $X_k = X(k)$ and $c_k = c(X_k)$.

Define a *linear* function approximator:

$$h^\theta(x) = \psi(x)^\top \theta, \qquad \psi : \mathcal{X} \to \mathbb{R}^d, \quad \theta \in \mathbb{R}^d.$$

The features are fixed; they may be nonlinear functions of the state.

Idea: The Bellman and Poisson equations usually cannot hold at every state within this approximation space. Start with a simple surrogate: minimize squared temporal differences along finitely many observed transitions.

Squared TD error for discounted cost

The Bellman equation is

$$h(X_k) = c_k + \gamma \mathbb{E}[h(X_{k+1}) \mid X_k].$$

Define $\psi_k = \psi(X_k)$ and the sampled temporal difference

$$\delta_k(\theta) = c_k + \gamma \psi_{k+1}^\top \theta - \psi_k^\top \theta = c_k - z_k^\top \theta, \qquad z_k = \psi_k - \gamma \psi_{k+1}.$$

Surrogate objective: Minimize the mean-squared TD error (MSTDE), estimated by

$$J_{\mathrm{TD}}(\theta) = \mathbb{E}[\delta_k(\theta)^2], \qquad \widehat{J}_T(\theta) = \frac{1}{T} \sum_{k=0}^{T-1} (c_k - z_k^\top \theta)^2.$$

The expectation uses the stationary law $\pi(dx)P(x, dy)$ of (X_k, X_{k+1}). For a stationary ergodic trajectory, $\widehat{J}_T(\theta) \to J_{\mathrm{TD}}(\theta)$ almost surely for each fixed θ. Independence of the observations is not required.

MSTDE algorithm for discounted cost

Algorithm

Given $X_0, \ldots, X_T$ and $c_0, \ldots, c_{T-1}$, form

$$\widehat{G}_T = \frac{1}{T} \sum_{k=0}^{T-1} z_k z_k^\top, \qquad \widehat{q}_T = \frac{1}{T} \sum_{k=0}^{T-1} z_k c_k.$$

Solve the normal equations $\widehat{G}_T \widehat{\theta}_T = \widehat{q}_T$. If $\widehat{G}_T \succ 0$, the minimizer is unique.

Remark

Ridge regularization minimizes $\widehat{J}_T(\theta) + \mu \|\theta\|^2$, with $\mu > 0$. The normal equations become $(\widehat{G}_T + \mu I)\widehat{\theta}_T = \widehat{q}_T$, ensuring a unique minimizer and improving conditioning, at the cost of biasing the estimate toward zero.

Centered squared TD error for average cost

The relative value function satisfies the Poisson equation

$$h(X_k) = c_k - \eta + \mathbb{E}[h(X_{k+1}) \mid X_k], \qquad \eta = \int_{\mathcal{X}} c(x)\,\pi(dx).$$

Set $z_k = \psi_k - \psi_{k+1}$ and define

$$\delta_k(\theta) = c_k - \eta - z_k^\top \theta, \qquad J_{\mathrm{TD}}(\theta) = \mathbb{E}[\delta_k(\theta)^2].$$

Estimate η from the trajectory and use the empirical objective

$$\widehat{\eta}_T = \frac{1}{T}\sum_{k=0}^{T-1} c_k, \qquad \widehat{J}_T(\theta) = \frac{1}{T}\sum_{k=0}^{T-1} (c_k - \widehat{\eta}_T - z_k^\top \theta)^2.$$

The expectation uses the stationary law $\pi(dx)P(x, dy)$ of (X_k, X_{k+1}). Under the same stationary ergodic sampling assumption, $\widehat{\eta}_T \to \eta$ and $\widehat{J}_T(\theta) \to J_{\mathrm{TD}}(\theta)$ almost surely for each fixed θ.

MSTDE algorithm for average cost

Algorithm

Given the trajectory and $\widehat{\eta}_T$, form

$$\widehat{G}_T = \frac{1}{T}\sum_{k=0}^{T-1} z_k z_k^\top, \qquad \widehat{q}_T = \frac{1}{T}\sum_{k=0}^{T-1} z_k (c_k - \widehat{\eta}_T).$$

Solve $\widehat{G}_T \widehat{\theta}_T = \widehat{q}_T$. If $\widehat{G}_T \succ 0$, the minimizer is unique.

Remark

Adding a constant to h^θ leaves every TD residual unchanged. Thus a constant feature makes $\widehat{G}_T$ singular; remove that direction or impose a normalization constraint such as $h^{\widehat{\theta}_T}(x_{\mathrm{ref}}) = 0$.

Squared TD error versus Bellman/Poisson error

To treat both cost criteria together, set $(a, \widetilde{c}_k) = (\gamma, c_k)$ for discounted cost and $(a, \widetilde{c}_k) = (1, c_k - \eta)$ for average cost. The Bellman/Poisson residual is the conditional mean

$$r_\theta(X_k) = \mathbb{E}[\delta_k(\theta) \mid X_k], \qquad \delta_k(\theta) = \widetilde{c}_k + a h^\theta(X_{k+1}) - h^\theta(X_k).$$

Its mean-square equation error is $J_{\mathrm{eq}}(\theta) = \mathbb{E}[r_\theta(X_k)^2]$. Conditional variance decomposition gives

$$\boxed{J_{\mathrm{TD}}(\theta) = J_{\mathrm{eq}}(\theta) + a^2 \mathbb{E}[\mathrm{Var}(h^\theta(X_{k+1}) \mid X_k)].}$$

Consequence: Least squares also penalizes variation of successor predictions. This extra term depends on θ, so the minimizers generally differ, even when the true value belongs to the feature span. For deterministic transitions, the variance term vanishes.

Double sampling for Bellman/Poisson error

Goal: Minimize $J_{\mathrm{eq}}(\theta) = \mathbb{E}[r_\theta(X)^2]$ instead. For linear features,

$$\nabla J_{\mathrm{eq}}(\theta) = 2\mathbb{E}[\mathbb{E}[\delta(\theta) \mid X]\, \mathbb{E}[\nabla_\theta \delta(\theta) \mid X]]\,.$$

Reusing one successor in both factors generally estimates the gradient of J_{TD}, not of J_{eq}.

Double sampling: Given the same state X, draw two conditionally independent successors Y_1, Y_2. With $\delta_i(\theta) = \widetilde{c}(X) + a h^\theta(Y_i) - h^\theta(X)$,

$$\mathbb{E}[\delta_1(\theta)\delta_2(\theta) \mid X] = r_\theta(X)^2.$$

Thus $2\delta_1(\theta)\nabla_\theta \delta_2(\theta)$ is an unbiased gradient estimator at fixed θ. In the average-cost case this identity uses the true η; estimating it introduces an additional error.

MSTDE – strengths and limitations

- **Simple computation:** Fixed linear features give a convex quadratic objective and a direct regression solve.
- **Wrong objective in general:** Sampled TD regression minimizes equation error plus a prediction-variance penalty.
- **Data and conditioning:** Conditioning can be bad; regularization changes the objective.
- **Double-sampling access:** Independent successors at the same state usually require a simulator that can restart there; an ordinary trajectory does not generally provide them.

A known transition model or estimates of conditional expectations offer other ways to approach Bellman/Poisson error minimization.

Table of Contents

Stochastic systems and value equations

Least-squares minimization of TD residuals

TD(0) – learn from one transition

Eligibility traces and TD(λ) – propagating information backward

Off-policy learning – exploration and stability

Learning a control policy – policy iteration, SARSA and Q-learning

Learning a parametrized policy – policy gradients and actor–critic

☕ Going further – modern reinforcement learning algorithms

From residual regression to TD(0)

With $h^\theta(x) = \psi(x)^\top \theta$, the discounted TD residual is

$$\delta_k(\theta) = c_k + \gamma h^\theta(X_{k+1}) - h^\theta(X_k).$$

Residual-gradient descent: Differentiate $\frac{1}{2}\delta_k(\theta)^2$ through both the current and successor predictions:

$$-\nabla_\theta \tfrac{1}{2}\delta_k(\theta)^2 = \delta_k(\theta)(\psi_k - \gamma\psi_{k+1}).$$

TD(0): First compute the bootstrap target $Y_k = c_k + \gamma h^{\theta_k}(X_{k+1})$. Hold Y_k fixed and differentiate only the current prediction:

$$\ell_k(\theta) = \tfrac{1}{2}(Y_k - h^\theta(X_k))^2,$$
$$-\nabla_\theta \ell_k(\theta)\big|_{\theta=\theta_k} = (Y_k - h^{\theta_k}(X_k))\psi_k = \delta_k(\theta_k)\psi_k.$$

This is a *semi-gradient*: the bootstrap target is recomputed at every step, but its dependence on the parameters is ignored during differentiation. TD(0) therefore does not perform gradient descent on MSTDE.

Some useful notation for the following

Let π be the invariant distribution and assume ψ belongs to $L^2(\pi)$.

Feature span:

$$\mathcal{H} = \{\psi^\top \theta : \theta \in \mathbb{R}^d\}.$$

Bellman operator:

$$\mathcal{T}_\gamma f = c + \gamma P f.$$

Poisson operator:

$$\mathcal{T}_{\text{av}} f = c - \eta + Pf, \qquad \eta = \int_{\mathcal{X}} c(x)\, \pi(dx).$$

Orthogonal projection: Π_π projects onto $\mathcal{H}$ in $L^2(\pi)$:

$$\Pi_\pi f = \arg\min_{v \in \mathcal{H}} \|f - v\|_\pi, \qquad \|f\|_\pi^2 = \int_{\mathcal{X}} f(x)^2\, \pi(dx).$$

TD(0) algorithm for discounted cost

Algorithm

Initialize θ_0 and choose step sizes $\alpha_k > 0$. At step k, observe transition (X_k, c_k, X_{k+1}) and update

$$\begin{aligned} Y_k &= c_k + \gamma h^{\theta_k}(X_{k+1}), \\ \delta_k &= Y_k - h^{\theta_k}(X_k), \\ \theta_{k+1} &= \theta_k + \alpha_k \delta_k \psi_k. \end{aligned}$$

TD(0) approximation target for discounted cost

For a fixed parameter θ, the stationary[1] mean update is

$$\mathbb{E}[\psi_k \delta_k(\theta)] = b - A_\gamma \theta,$$

where

$$A_\gamma = \int_{\mathcal{X}} \psi(x)\big(\psi(x) - \gamma(P\psi)(x)\big)^\top \pi(dx),$$
$$b = \int_{\mathcal{X}} \psi(x) c(x)\, \pi(dx).$$

Approximation target: Equilibrium of stationary mean update:

$$\mathbb{E}[\psi_k \delta_k(\theta)] = 0 \quad \Longleftrightarrow \quad A_\gamma \theta = b$$
$$\Longleftrightarrow \quad \mathcal{T}_\gamma h^\theta - h^\theta \ \perp \ \mathcal{H}.$$

[1] Using the stationary law $\pi(dx)P(x, dy)$ on (X_k, X_{k+1}).

Projected Bellman equation and approximation error

Set $C = \int_{\mathcal{X}} \psi(x)\psi(x)^\top \pi(dx)$.

Proposition

If c and the features ψ belong to $L^2(\pi)$ and $C \succ 0$, the equation

$$\boxed{h^{\theta_\gamma} = \Pi_\pi \mathcal{T}_\gamma h^{\theta_\gamma}}$$

has a unique solution, characterized by $A_\gamma \theta_\gamma = b$. The mean dynamics $\dot{\theta} = b - A_\gamma \theta$ converge globally exponentially to θ_γ. Moreover,

$$\|h^{\theta_\gamma} - h\|_\pi \leq \frac{\|\Pi_\pi h - h\|_\pi}{\sqrt{1-\gamma^2}}.$$

The TD limiting approximation need not minimize mean-square value error or Bellman error. The bound compares it with the best value approximation in the feature span.

Projected Bellman equation: proof

Invariance and conditional Jensen imply $\|Pf\|_\pi \le \|f\|_\pi$. Hence $\Pi_\pi \mathcal{T}_\gamma$ is a γ-contraction on the feature span and has a unique fixed point. Orthogonality of its residual is exactly $A_\gamma \theta = b$.

Since $h = \mathcal{T}_\gamma h$, orthogonality gives

$$\begin{aligned}\|h^{\theta_\gamma} - h\|_\pi^2 &= \|h^{\theta_\gamma} - \Pi_\pi h\|_\pi^2 + \|\Pi_\pi h - h\|_\pi^2 \\ &\le \gamma^2 \|h^{\theta_\gamma} - h\|_\pi^2 + \|\Pi_\pi h - h\|_\pi^2.\end{aligned}$$

Rearranging proves the bound. For $f_z = \psi^\top z$, Cauchy–Schwarz gives

$$\int_{\mathcal{X}} f_z(x)(Pf_z)(x)\,\pi(dx) \le \|f_z\|_\pi \|Pf_z\|_\pi \le \|f_z\|_\pi^2.$$

Hence

$$z^\top A_\gamma z = \|f_z\|_\pi^2 - \gamma \int_{\mathcal{X}} f_z(x)(Pf_z)(x)\,\pi(dx) \ge (1-\gamma) z^\top C z.$$

Thus the mean dynamics $\dot\theta = b - A_\gamma \theta$ are globally stable.

TD(0) algorithm for average cost

Idea: Center the cost using the current estimate ρ_k.

Algorithm

Initialize θ_0, ρ_0 and choose step sizes $\alpha_k > 0$. At step k, observe transition (X_k, c_k, X_{k+1}) and update

$$\begin{aligned}
\delta_k &= c_k - \rho_k + h^{\theta_k}(X_{k+1}) - h^{\theta_k}(X_k), \\
\rho_{k+1} &= \rho_k + \alpha_k(c_k - \rho_k), \\
\theta_{k+1} &= \theta_k + \alpha_k \delta_k \psi_k.
\end{aligned}$$

Remark

Use linearly independent features whose span contains no nonzero constant function. Otherwise the value parameters have an unidentifiable constant direction.

TD(0) approximation target for average cost

For fixed parameters (θ, ρ), the stationary mean updates are

$$\begin{aligned}\mathbb{E}[c_k - \rho] &= \eta - \rho, \\ \mathbb{E}[\psi_k \delta_k(\theta, \rho)] &= b_{\mathrm{av}} - A_{\mathrm{av}}\theta + (\eta - \rho)m,\end{aligned}$$

where $m = \int_{\mathcal{X}} \psi(x)\, \pi(dx)$ and

$$\begin{aligned}A_{\mathrm{av}} &= \int_{\mathcal{X}} \psi(x)\big(\psi(x) - (P\psi)(x)\big)^{\top} \pi(dx), \\ b_{\mathrm{av}} &= \int_{\mathcal{X}} \psi(x)(c(x) - \eta)\, \pi(dx).\end{aligned}$$

Approximation target: Equilibrium of the stationary mean updates:

$$\begin{cases}\mathbb{E}[c_k - \rho] = 0, \\ \mathbb{E}[\psi_k \delta_k(\theta, \rho)] = 0\end{cases} \quad \Longleftrightarrow \quad \begin{cases}\rho = \eta, \\ A_{\mathrm{av}}\theta = b_{\mathrm{av}}.\end{cases}$$

Projected Poisson equation and stability

The equilibrium equation for the value parameters is equivalent to

$$A_{\mathrm{av}}\theta = b_{\mathrm{av}} \quad \Longleftrightarrow \quad \mathcal{T}_{\mathrm{av}} h^{\theta} - h^{\theta} \perp \mathcal{H}.$$

For $f_z = \psi^{\top} z$, stationarity gives

$$z^{\top} A_{\mathrm{av}} z = \frac{1}{2}\mathbb{E}[(f_z(X_{k+1}) - f_z(X_k))^2].$$

Proposition

If c and the features ψ belong to $L^2(\pi)$ and every nonzero f_z changes across a stationary transition with positive probability, the equation

$$\boxed{h^{\theta_{\mathrm{av}}} = \Pi_{\pi} \mathcal{T}_{\mathrm{av}} h^{\theta_{\mathrm{av}}}}$$

has a unique solution, characterized by $A_{\mathrm{av}}\theta_{\mathrm{av}} = b_{\mathrm{av}}$. The joint mean dynamics $\dot{\rho} = \eta - \rho$ and $\dot{\theta} = b_{\mathrm{av}} - A_{\mathrm{av}}\theta + (\eta - \rho)m$ converge globally exponentially to $(\eta, \theta_{\mathrm{av}})$.

TD(0) – sufficient convergence conditions

A sufficient setting for both TD(0) algorithms is:

- Uniform geometric ergodicity with invariant distribution π
- Bounded costs and features, with $C \succ 0$
- Step sizes $\alpha_k = s/(k+1)^\kappa$, where $s > 0$ and $1/2 < \kappa \leq 1$
- For average cost, the positive-definiteness condition on the preceding slide

Theorem

Under these conditions: discounted TD(0) satisfies $\theta_k \to \theta_\gamma$ almost surely; average-cost TD(0) satisfies $\rho_k \to \eta$ and $\theta_k \to \theta_{\mathrm{av}}$ almost surely.

Tsitsiklis & Van Roy (1997), discounted TD; Tsitsiklis & Van Roy (1999), average-cost TD.

TD(0) convergence: stochastic-approximation result

We use the following standard result for linear updates with Markov noise.

Theorem (Linear stochastic approximation)

Consider

$$z_{k+1} = z_k + \alpha_k \big(M(Y_k) z_k + d(Y_k)\big),$$

where Y_k is a time-homogeneous, uniformly geometrically ergodic Markov process, and M, d are bounded measurable functions. Let $\overline{M}, \overline{d}$ be their invariant expectations. If $\overline{M}$ is Hurwitz and $\alpha_k = s/(k+1)^\kappa$, with $s > 0$ and $1/2 < \kappa \leq 1$, then, from any finite initial vector,

$$z_k \longrightarrow -\overline{M}^{-1}\overline{d} \quad \textit{almost surely.}$$

Standard linear-SA result; see Tsitsiklis & Van Roy (1999), Section 3.3, Theorem 2.

TD(0) convergence for discounted cost: proof

Set $Y_k = (X_k, X_{k+1})$. The transition-pair process is Markov and inherits uniform geometric ergodicity from X_k after its first transition. Its invariant law is $\pi(dx)P(x, dy)$.

The discounted TD update has the required linear form, with

$$M(Y_k) = \psi_k(\gamma\psi_{k+1} - \psi_k)^\top, \qquad d(Y_k) = \psi_k c_k.$$

Bounded costs and features make M, d bounded, and

$$\overline{M} = -A_\gamma, \qquad \overline{d} = b.$$

The feature condition $C \succ 0$ makes $-A_\gamma$ Hurwitz. The stochastic-approximation result therefore yields

$$\theta_k \longrightarrow A_\gamma^{-1} b = \theta_\gamma \quad \text{almost surely.}$$

TD(0) convergence for average cost: proof

With the same Y_k, stack the parameters as $z_k = (\rho_k, \theta_k^\top)^\top$. The joint update has the linear coefficients

$$M(Y_k) = \begin{pmatrix} -1 & 0 \\ -\psi_k & \psi_k(\psi_{k+1} - \psi_k)^\top \end{pmatrix}, \qquad d(Y_k) = \begin{pmatrix} c_k \\ \psi_k c_k \end{pmatrix}.$$

These are bounded, with invariant expectations

$$\overline{M} = \begin{pmatrix} -1 & 0 \\ -m & -A_{\mathrm{av}} \end{pmatrix}, \qquad \overline{d} = \begin{pmatrix} \eta \\ b_{\mathrm{av}} + \eta m \end{pmatrix}.$$

The feature-variation condition makes $-A_{\mathrm{av}}$ Hurwitz. Since $\overline{M}$ is block triangular, it is also Hurwitz. Solving $\overline{M}z + \overline{d} = 0$ gives $\rho = \eta$ and $A_{\mathrm{av}}\theta = b_{\mathrm{av}}$. The stochastic-approximation result therefore yields

$$(\rho_k, \theta_k) \longrightarrow (\eta, \theta_{\mathrm{av}}) \quad \text{almost surely.}$$

TD(0) – strengths and limitations

- **Online and inexpensive:** One transition, $O(d)$ work and $O(d)$ parameter storage
- **A different approximation target:** Projected Bellman/Poisson equations, rather than least-squares minimization of sampled residuals
- **Step-size choice:** Diminishing steps support exact asymptotic convergence; constant steps generally leave persistent fluctuations
- **Qualified guarantees:** The stated theory uses fixed linear features and on-policy sampling

Nonlinear parametrizations, changing policies and off-policy sampling need separate analysis; the same update formula does not guarantee stability.

Table of Contents

Stochastic systems and value equations

Least-squares minimization of TD residuals

TD(0) – learn from one transition

Eligibility traces and TD(λ) – propagating information backward

Off-policy learning – exploration and stability

Learning a control policy – policy iteration, SARSA and Q-learning

Learning a parametrized policy – policy gradients and actor–critic

☕ Going further – modern reinforcement learning algorithms

Eligibility traces: updating earlier states

Note: We present TD(1) only for the discounted cost.

TD(0) uses $\delta_k \psi_k$: a new error directly updates only the current feature vector. Can the same error also improve predictions at recently visited states?

Eligibility trace: Keep a discounted memory of past features:

$$e_k = \psi_k + \gamma \psi_{k-1} + \cdots + \gamma^k \psi_0 = \gamma e_{k-1} + \psi_k.$$

With the update $\theta_{k+1} = \theta_k + \alpha_k \delta_k e_k$, an error at time k contributes in the direction

$$\delta_k \psi_k, \quad \gamma \delta_k \psi_{k-1}, \quad \gamma^2 \delta_k \psi_{k-2}, \quad \ldots$$

The trace combines feature vectors of previously visited states, so a new TD error updates the parameters along those past feature directions. In a tabular representation, a delayed cost can thus affect several previously visited states in one update.

TD(1) algorithm

Algorithm

Initialize θ_0 and $e_{-1} = 0$. At step k, observe transition (X_k, c_k, X_{k+1}) and update

$$\begin{aligned} e_k &= \gamma e_{k-1} + \psi_k, \\ \delta_k &= c_k + \gamma h^{\theta_k}(X_{k+1}) - h^{\theta_k}(X_k), \\ \theta_{k+1} &= \theta_k + \alpha_k \delta_k e_k. \end{aligned}$$

Remark

Compared with TD(0), replace ψ_k by the trace e_k; the one-step TD error δ_k is unchanged.

TD(1) and the full discounted return

Define $G_k = \sum_{j=0}^{\infty} \gamma^j c_{k+j}$. For bounded costs, $\mathbb{E}[G_k \mid X_k] = h(X_k)$. A return-regression update would use $(G_k - h^\theta(X_k))\psi_k$.

Why TD errors contain the return: Freeze θ and telescope:

$$\sum_{j=0}^{n-1} \gamma^j \delta_{k+j}(\theta) = \sum_{j=0}^{n-1} \gamma^j c_{k+j} + \gamma^n h^\theta(X_{k+n}) - h^\theta(X_k).$$

If the features are bounded, the terminal term vanishes, giving

$$G_k - h^\theta(X_k) = \sum_{j=0}^{\infty} \gamma^j \delta_{k+j}(\theta).$$

Traces assign these errors backward as they arrive, without waiting to observe G_k. The identity freezes θ; conventional online TD(1), which changes θ_k at every step, is not exactly an online implementation of fixed-target return regression.

TD(1) approximation target: best value approximation

Use the stationary trace $e_k = \sum_{j=0}^{\infty} \gamma^j \psi_{k-j}$ on a two-sided stationary trajectory. Assume bounded costs and features, and define

$$C = \int_{\mathcal{X}} \psi(x)\psi(x)^\top \pi(dx), \qquad b_1 = \int_{\mathcal{X}} \psi(x)h(x)\,\pi(dx).$$

Proposition

The stationary mean TD(1) update at fixed θ is

$$\mathbb{E}[e_k \delta_k(\theta)] = b_1 - C\theta.$$

If $C \succ 0$, its unique equilibrium, characterized by $C\theta_{\gamma,1} = b_1$, satisfies

$$\boxed{h^{\theta_{\gamma,1}} = \Pi_\pi h.}$$

Thus TD(1) targets the minimizer of $\|h^\theta - h\|_\pi^2$, and its mean dynamics $\dot{\theta} = b_1 - C\theta$ converge globally exponentially to $\theta_{\gamma,1}$.

TD(1) approximation target: proof

For fixed θ, boundedness and $\gamma < 1$ justify exchanging sums and expectations. Stationarity shifts a past feature paired with the current error into a current feature paired with a future error:

$$\begin{aligned}
\mathbb{E}[e_k \delta_k(\theta)] &= \sum_{j=0}^{\infty} \gamma^j \mathbb{E}[\psi_{k-j} \delta_k(\theta)] \\
&= \sum_{j=0}^{\infty} \gamma^j \mathbb{E}[\psi_k \delta_{k+j}(\theta)] \\
&= \mathbb{E}[\psi_k (G_k - h^\theta(X_k))] \\
&= \int_{\mathcal{X}} \psi(x)(h(x) - h^\theta(x))\, \pi(dx) = b_1 - C\theta.
\end{aligned}$$

Since $C \succ 0$, the equilibrium is unique and $\dot{\theta} = b_1 - C\theta$ is globally exponentially stable.

This identifies the limiting approximation; online convergence additionally requires control of the trace-dependent noise.

TD(1) – sufficient convergence conditions

Assume the same setting as for discounted TD(0):

- Uniform geometric ergodicity with invariant distribution π
- Bounded costs and features, with $C \succ 0$
- Step sizes $\alpha_k = s/(k+1)^\kappa$, where $s > 0$ and $1/2 < \kappa \leq 1$

Theorem

Under these conditions, discounted TD(1) satisfies $\theta_k \to \theta_{\gamma,1}$ almost surely.

Tsitsiklis & Van Roy (1997), Theorem 1 and trace-process analysis.

TD(1) convergence: proof (1/2)

Let $g(x, y)$ be bounded. For $0 \leq j \leq n$, define

$$H_j(x) = \mathbb{E}[\psi(X_0) g(X_j, X_{j+1}) \mid X_0 = x],$$
$$\overline{H}_j = \int_{\mathcal{X}} \int_{\mathcal{X}} \int_{\mathcal{X}} \psi(x) g(y, z)\, P(y, dz)\, P^j(x, dy)\, \pi(dx).$$

Uniform geometric ergodicity yields

$$\|P^m H_j(x) - \overline{H}_j\| \leq K r^m \quad \text{for all } x,\ 0 \leq j,\ 0 \leq m.$$

Let $Y_k = (X_k, X_{k+1}, e_k)$ and $F(Y_k) = e_k g(X_k, X_{k+1})$.

For $0 \leq j < n$ and $y_0 = (x_0, x_1, \bar{e}_0)$, conditioning on $X_1 = x_1$ gives

$$\mathbb{E}[\psi(X_{n-j}) g(X_n, X_{n+1}) \mid Y_0 = y_0] = P^{n-j-1} H_j(x_1).$$

TD(1) convergence for discounted cost: proof (2/2)

Expanding the trace $e_k = \sum_{j=0}^{k} \gamma^j \psi(X_{k-j})$ and using the above gives

$$\begin{aligned}&\mathbb{E}[F(Y_n) \mid Y_0 = y_0] \\ &\quad = \gamma^n \bar{e}_0 \mathbb{E}[g(X_n, X_{n+1}) \mid Y_0 = y_0] + \sum_{j=0}^{n-1} \gamma^j P^{n-j-1} H_j(x_1).\end{aligned}$$

With $\overline{F} = \sum_{j=0}^{\infty} \gamma^j \overline{H}_j$, the convergence from the preceding slide gives

$$\|\mathbb{E}[F(Y_n) \mid Y_0 = y_0] - \overline{F}\| \le K\Big(\gamma^n + \sum_{j=0}^{n-1} \gamma^j r^{n-1-j}\Big).$$

Apply this bound to M with $g(x, y) = (\gamma\psi(y) - \psi(x))^\top$, and to d with $g(x, y) = c(x)$. Linear stochastic approximation now yields $\theta_k \to C^{-1} b_1 = \theta_{\gamma,1}$ almost surely.

TD(0) versus TD(1) – learning speed and variability

- **Information propagation:** TD(1) lets a delayed error affect recently visited states immediately; TD(0) often needs later visits to propagate that information backward.
- **Dependence on predictions:** TD(0) uses a one-step bootstrap; TD(1) has the full-return mean-square approximation target.
- **Variability:** Full returns include more future randomness and often have higher variance than one-step bootstrap targets. Large traces can also magnify individual updates, especially when γ is close to 1.
- **Asymptotic approximation:** TD(1) reaches the best $L^2(\pi)$ value approximation under the linear convergence assumptions; TD(0) generally reaches a different projected fixed point.

Faster propagation does not guarantee faster convergence or lower finite-data error. The variance ordering is not universal, and the step size must be tuned together with the trace.

Average cost – why TD(1) needs separate treatment

For average cost, the natural centered error and trace are

$$\delta_k = c_k - \rho_k + h^{\theta_k}(X_{k+1}) - h^{\theta_k}(X_k),$$
$$e_k = e_{k-1} + \psi_k.$$

The trace accumulates all past features without decay and need not remain bounded.

Unlike discounted TD(1), there is no discount-induced trace decay. The infinite centered return $\sum_{j\geq 0}(c_{k+j} - \eta)$ need not converge along sample paths, even when the relative value is well defined through expected partial sums.

Thus neither the discounted TD(1) proof nor its Monte Carlo interpretation transfers by setting $\gamma = 1$. Regenerative returns with appropriate return-time moments and normalization provide one alternative.

TD(λ) – interpolate between one step and full return

Freeze θ and define the n-step return target

$$G_k^{(n)} = \sum_{j=0}^{n-1} \gamma^j c_{k+j} + \gamma^n h^\theta(X_{k+n}).$$

For $0 \leq \lambda < 1$, take a geometric mixture:

$$G_k^\lambda = (1-\lambda) \sum_{n=1}^{\infty} \lambda^{n-1} G_k^{(n)}.$$

With bounded costs and features, $G_k^\lambda \to G_k$ as $\lambda \uparrow 1$.

TD(λ) algorithm for discounted cost

Fix $0 \leq \lambda \leq 1$.

Algorithm

Initialize θ_0 and $e_{-1} = 0$. At step k, observe transition (X_k, c_k, X_{k+1}) and update

$$\begin{aligned} e_k &= \gamma\lambda e_{k-1} + \psi_k, \\ \delta_k &= c_k + \gamma h^{\theta_k}(X_{k+1}) - h^{\theta_k}(X_k), \\ \theta_{k+1} &= \theta_k + \alpha_k \delta_k e_k. \end{aligned}$$

TD(λ) algorithm for average cost

Fix $0 \leq \lambda < 1$.

Algorithm

Initialize θ_0, ρ_0 and $e_{-1} = 0$. At step k, observe transition (X_k, c_k, X_{k+1}) and update

$$\begin{aligned} e_k &= \lambda e_{k-1} + \psi_k, \\ \delta_k &= c_k - \rho_k + h^{\theta_k}(X_{k+1}) - h^{\theta_k}(X_k), \\ \rho_{k+1} &= \rho_k + \alpha_k(c_k - \rho_k), \\ \theta_{k+1} &= \theta_k + \alpha_k \delta_k e_k. \end{aligned}$$

Remark

Use linearly independent features with no nonzero constant in $\mathcal{H}$.

TD(λ) approximation target for discounted cost

For $0 \leq \lambda < 1$, define the multistep Bellman operator

$$\mathcal{T}_{\gamma,\lambda} f = (1-\lambda) \sum_{n=1}^{\infty} \lambda^{n-1} \mathcal{T}_{\gamma}^{n} f.$$

Approximation target: With $C \succ 0$, the stationary mean update vanishes at the unique solution of

$$\boxed{h^{\theta_{\gamma,\lambda}} = \Pi_{\pi} \mathcal{T}_{\gamma,\lambda} h^{\theta_{\gamma,\lambda}}.}$$

Tsitsiklis & Van Roy (1997), Sections III–IV, Theorem 1.

Remark

At $\lambda = 0$, this is the projected Bellman equation. As $\lambda \uparrow 1$, $\mathcal{T}_{\gamma,\lambda} f \to h$, so this is consitent with TD(1) limiting approximation $h^{\theta_{\gamma,1}} = \Pi_{\pi} h$.

TD(λ) approximation target for average cost

For $0 \leq \lambda < 1$, define the multistep centered Poisson operator

$$\mathcal{T}_{\mathrm{av},\lambda} f = (1-\lambda) \sum_{n=1}^{\infty} \lambda^{n-1} \mathcal{T}_{\mathrm{av}}^{n} f,$$

Approximation target: Under the positive-definiteness condition on the next slide, the stationary mean updates vanish at $\rho = \eta$ and the unique solution of

$$\boxed{h^{\theta_{\mathrm{av},\lambda}} = \Pi_{\pi} \mathcal{T}_{\mathrm{av},\lambda} h^{\theta_{\mathrm{av},\lambda}}.}$$

Tsitsiklis & Van Roy (1999), Section 3, Theorem 1.

Remark

At $\lambda = 0$, it is the projected Poisson equation. As $\lambda \uparrow 1$, any limiting operator can retain an additive constant depending on f; it need not return the normalized relative value h.

TD(λ) – sufficient convergence conditions

A sufficient setting for both TD(λ) algorithms is:

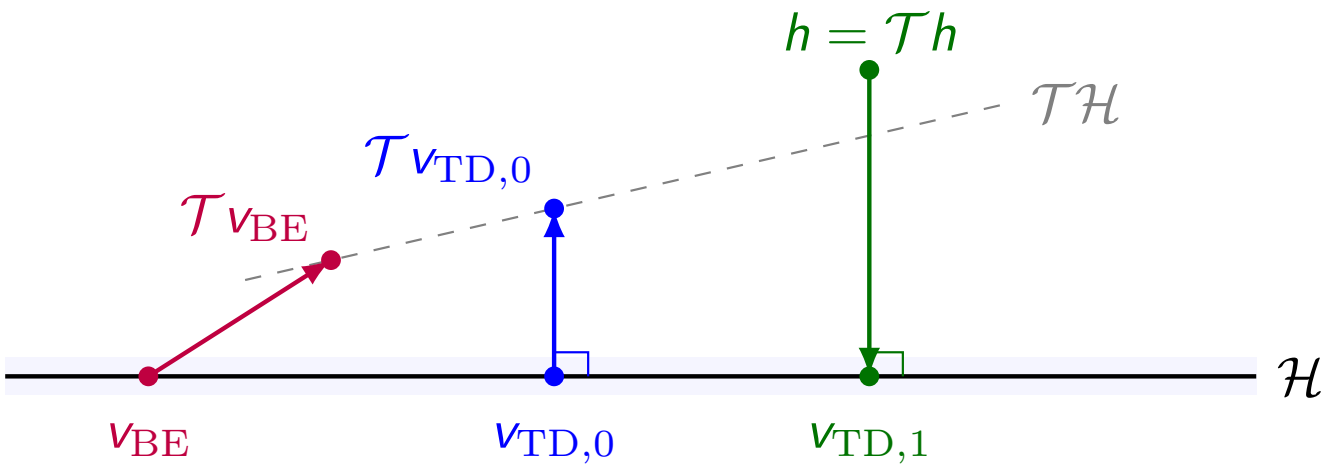
- Uniform geometric ergodicity with invariant distribution π
- Bounded costs and features, with $C \succ 0$
- Step sizes $\alpha_k = s/(k+1)^\kappa$, where $s > 0$ and $1/2 < \kappa \leq 1$
- For average cost, the positive-definiteness condition stated for TD(0)

Theorem

Under these conditions: discounted TD(λ), for each fixed $0 \leq \lambda \leq 1$, satisfies $\theta_k \to \theta_{\gamma,\lambda}$ almost surely; average-cost TD(λ), for each fixed $0 \leq \lambda < 1$, satisfies $\rho_k \to \eta$ and $\theta_k \to \theta_{\mathrm{av},\lambda}$ almost surely.

Tsitsiklis & Van Roy (1997), Theorem 1; Tsitsiklis & Van Roy (1999), Section 3.3, Theorem 2 and Corollary 1, with the trace-noise argument above.

Comparison of approximation targets



$$v_{\mathrm{BE}} \in \arg\min_{v \in \mathcal{H}} \|\mathcal{T}v - v\|_\pi^2, \qquad \text{minimum Bellman error,}$$

$$v_{\mathrm{TD},0} = \Pi_\pi \mathcal{T} v_{\mathrm{TD},0}, \qquad \text{TD(0) fixed point,}$$

$$v_{\mathrm{TD},1} = \arg\min_{v \in \mathcal{H}} \|v - h\|_\pi^2, \qquad \text{TD(1) approximation target.}$$

Remember that $v_{\mathrm{BE}} \neq v_{\mathrm{MSTDE}}$.

Illustration inspired by Sutton & Barto (2018), Section 11.4, Figure 11.4.

LSTD(λ) for discounted cost

Idea: Replace incremental TD updates by a linear solve that makes their empirical mean zero. Fix $0 \le \lambda \le 1$ and set $e_{-1} = 0$.

Algorithm

At step k, observe transition (X_k, c_k, X_{k+1}) and accumulate

$$e_k = \gamma\lambda e_{k-1} + \psi_k,$$

$$\widehat{A}_T = \frac{1}{T}\sum_{k=0}^{T-1} e_k(\psi_k - \gamma\psi_{k+1})^\top,$$

$$\widehat{b}_T = \frac{1}{T}\sum_{k=0}^{T-1} e_k c_k.$$

After T transitions, solve $\widehat{A}_T\widehat{\theta}_T = \widehat{b}_T$ when $\widehat{A}_T$ is nonsingular.

LSTD(λ) for average cost

Fix $0 \le \lambda < 1$ and set $e_{-1} = 0$.

Algorithm

At step k, observe transition (X_k, c_k, X_{k+1}) and accumulate

$$
\begin{aligned}
e_k &= \lambda e_{k-1} + \psi_k, \\
\widehat{\eta}_T &= \frac{1}{T} \sum_{k=0}^{T-1} c_k, \\
\widehat{A}_T &= \frac{1}{T} \sum_{k=0}^{T-1} e_k (\psi_k - \psi_{k+1})^\top, \\
\widehat{b}_T &= \frac{1}{T} \sum_{k=0}^{T-1} e_k (c_k - \widehat{\eta}_T).
\end{aligned}
$$

Solve $\widehat{A}_T \widehat{\theta}_T = \widehat{b}_T$ when $\widehat{A}_T$ is nonsingular.

LSTD(λ) – approximation target and consistency

Same approximation target as TD(λ):

$$A_\lambda \theta_\lambda = b_\lambda \quad \Longleftrightarrow \quad h^{\theta_\lambda} = \Pi_\pi \mathcal{T}_\lambda h^{\theta_\lambda},$$

where $\mathcal{T}_\lambda$ denotes $\mathcal{T}_{\gamma,\lambda}$ or $\mathcal{T}_{\mathrm{av},\lambda}$.

Theorem

Under the boundedness and ergodicity conditions stated for TD(λ),

$$\widehat{A}_T \to A_\lambda, \qquad \widehat{b}_T \to b_\lambda, \qquad \widehat{\theta}_T \to \theta_\lambda, \qquad \textit{almost surely.}$$

The empirical matrix is nonsingular for all sufficiently large T. For average cost, also $\widehat{\eta}_T \to \eta$ almost surely.

No step-size schedule is needed for these batch estimates.

LSTD(λ) versus TD(λ)

- **Same approximation target:** Both target the same projected equation
- **Different computation:** TD uses $O(d)$ work and storage per update; dense LSTD accumulates $d \times d$ statistics with $O(d^2)$ work per transition and storage
- **Linear solve:** A dense solve costs $O(d^3)$; solve periodically or use recursive updates with appropriate numerical safeguards
- **Finite data:** The empirical matrix can be singular or poorly conditioned, even when the population matrix is nonsingular
- **Tuning:** LSTD removes the TD step size, but feature choice, λ, data coverage and regularization still matter

$\widehat{A}_T$ is generally nonsymmetric: it is not the Gram matrix from MSTDE regression. Regularization changes the empirical equation and can bias the limiting approximation if kept fixed.

Nonlinear approximation – discounted TD(λ)

Evaluate a fixed policy using a differentiable value model $h^\theta(x)$, e.g., a neural network, with $\theta \in \mathbb{R}^d$. Replace fixed features by

$$g_k \coloneqq \nabla_\theta h^\theta(X_k)\big|_{\theta=\theta_k}.$$

Algorithm

Fix $0 < \gamma < 1$, $0 \le \lambda \le 1$, and initialize $e_{-1} = 0$. From each on-policy transition (X_k, c_k, X_{k+1}), compute

$$\begin{aligned} e_k &= \gamma\lambda e_{k-1} + g_k, \\ \delta_k &= c_k + \gamma h^{\theta_k}(X_{k+1}) - h^{\theta_k}(X_k), \\ \theta_{k+1} &= \theta_k + \alpha_k \delta_k e_k. \end{aligned}$$

For $h^\theta(x) = \psi(x)^\top\theta$, $g_k = \psi(X_k)$ and we recover linear TD(λ).

Sutton & Barto, Section 12.2: semi-gradient TD(λ).

Nonlinear TD – semi-gradients and changing traces

At $\lambda = 0$, freeze the bootstrap target $Y_k = c_k + \gamma h^{\theta_k}(X_{k+1})$ during differentiation:

$$-\nabla_\theta \tfrac{1}{2}(Y_k - h^\theta(X_k))^2\big|_{\theta=\theta_k} = \delta_k g_k.$$

Traces remember past gradients:

$$e_k = \sum_{j=0}^{k} (\gamma\lambda)^{k-j} \nabla_\theta h^\theta(X_j)\big|_{\theta=\theta_j}.$$

These gradients were evaluated at different parameters θ_j.
Frozen-parameter forward-view identities therefore do not give exact equivalence to conventional online updates, even at $\lambda = 1$.

Nonlinear approximation – average-cost TD(λ)

For a fixed policy, approximate the relative value by h^θ and the average cost by ρ. A direct semi-gradient extension, with $0 \le \lambda < 1$, is

$$
\begin{aligned}
g_k &= \nabla_\theta h^\theta(X_k)\big|_{\theta=\theta_k}, \\
e_k &= \lambda e_{k-1} + g_k, \\
\delta_k &= c_k - \rho_k + h^{\theta_k}(X_{k+1}) - h^{\theta_k}(X_k), \\
\rho_{k+1} &= \rho_k + \alpha_k(c_k - \rho_k), \\
\theta_{k+1} &= \theta_k + \alpha_k \delta_k e_k.
\end{aligned}
$$

Normalization: Remove the arbitrary additive constant, e.g., parameterize

$$h^\theta(x) = f^\theta(x) - f^\theta(x_{\mathrm{ref}}),$$

and differentiate this entire expression when forming g_k.

This defines an algorithm, not a convergence theorem: the preceding linear average-cost result does not apply to it automatically.

Nonlinear TD(λ) – strengths and limitations

- **Flexible representation:** Learn features together with values; automatic differentiation supplies g_k, and the trace stores one number per parameter
- **No fixed linear system:** The gradients depend on θ: the mean update is no longer $b_\lambda - A_\lambda \theta$, and ordinary LSTD is no longer a single solve for all parameters
- **No inherited convergence guarantee:** The fixed-subspace projection argument no longer applies. Nonlinear bootstrapped TD can be unstable even on-policy
- **Return regression is also nonconvex:** Fitting complete returns removes the bootstrap target, but nonlinear least squares can still have multiple stationary points

The benefit is a richer value model; stability and approximation quality require assumptions on both the model and the learning dynamics.

Table of Contents

Stochastic systems and value equations

Least-squares minimization of TD residuals

TD(0) – learn from one transition

Eligibility traces and TD(λ) – propagating information backward

Off-policy learning – exploration and stability

Learning a control policy – policy iteration, SARSA and Q-learning

Learning a parametrized policy – policy gradients and actor–critic

☕ Going further – modern reinforcement learning algorithms

TD also applies to Q-functions

Fix a stationary policy ϕ and consider discounted cost. Under ϕ, the state–action pair (X_k, U_k) is a Markov process with value Q_ϕ. Use fixed features $\psi(x, u)$ and $Q^\theta(x, u) = \psi(x, u)^\top \theta$.

On-policy evaluation: Execute ϕ, including $U_{k+1} \sim \phi(\cdot \mid X_{k+1})$, and update

$$\delta_k = c_k + \gamma Q^{\theta_k}(X_{k+1}, U_{k+1}) - Q^{\theta_k}(X_k, U_k),$$
$$\theta_{k+1} = \theta_k + \alpha_k \delta_k \psi(X_k, U_k).$$

The linear TD evaluation guarantees transfer if their assumptions hold for the state–action chain.

A deterministic policy does not sample alternative actions. Learning Q-values for policy improvement requires sufficient state–action coverage.

Off-policy bootstrap targets and exploration

Two policies: The behavior policy b generates data (X_k, U_k); the target policy ϕ is being evaluated. The TD residual is

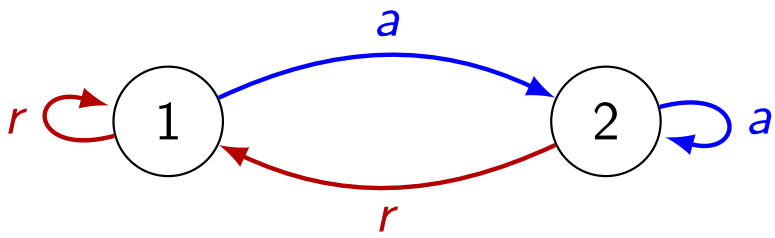
$$\delta_k = c_k + \gamma \int_{\mathcal{U}} Q^{\theta_k}(X_{k+1}, u)\, \phi(du \mid X_{k+1}) - Q^{\theta_k}(X_k, U_k).$$

Using the actual next action U_{k+1} from b instead of the integral would evaluate b instead of ϕ. The bootstrap target averages next-state predictions under ϕ, but the frequency of updates is determined by b.

Exploration: For action-value control, cover the state–action pairs (X_k, U_k) needed for improvement. A greedy behavior can miss useful actions; ϵ-greedy randomizes actions but does not by itself ensure visits to every state.

Q-learning: Replace the integral by $\min_a Q^{\theta_k}(X_{k+1}, a)$. This targets greedy control while behavior may remain exploratory. With function approximation, **convergence is not automatic**.

A two-state counterexample: divergent off-policy TD(0)



$h^\theta(1) = \theta$ $\quad$ $h^\theta(2) = 2\theta$

$c = 0$, $\gamma = 0.9$.
Target policy: $\phi(a \mid x) = 1$.
Behavior: $b(a \mid x) = 1/4$,
$b(r \mid x) = 3/4$.

Behavior invariant distribution: $(3/4, 1/4)$. The TD residual is

$$\delta_k(\theta) = \gamma \int_{\mathcal{U}} h^\theta(F(X_k, u))\, \phi(du \mid X_k) - h^\theta(X_k) = 1.8\theta - h^\theta(X_k).$$

At fixed θ, the stationary mean update is

$$\mathbb{E}_b[\delta_k(\theta)\psi(X_k)] = \tfrac{3}{4}(0.8\theta) + \tfrac{1}{4}(-0.4\theta) = \tfrac{1}{2}\theta.$$

Thus $\dot\theta = \theta/2$ is unstable, although the true value $h = 0$ is representable.

The deadly triad

Instability can arise when these three ingredients occur together:

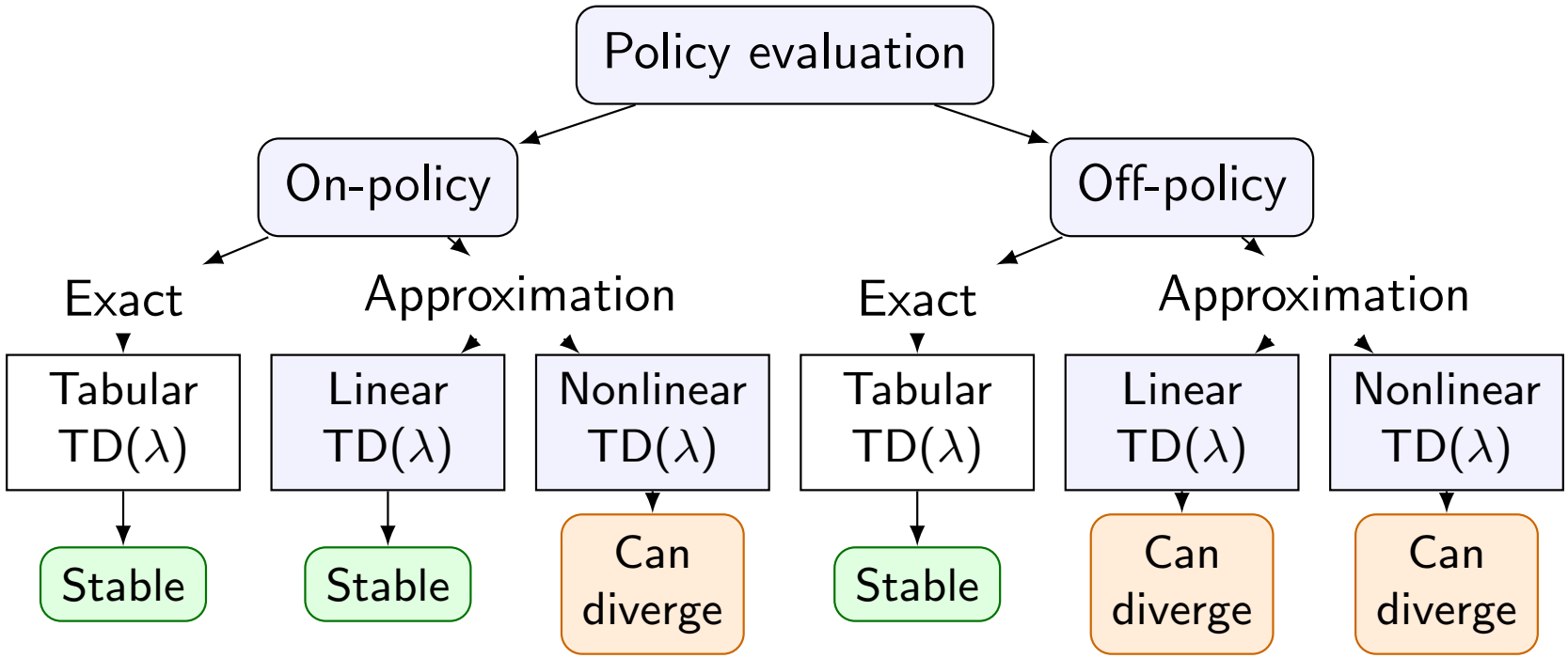
1. **Function approximation:** Parameters couple predictions at different states or actions
2. **Bootstrapping:** Bootstrap targets depend on current predictions
3. **Off-policy learning:** Data follow a different distribution from the target policy

Why earlier proofs fail: The target-policy transition operator need not be nonexpansive in the behavior-weighted norm. The projected operator can therefore lose its contraction property.

Possible responses: Use on-policy evaluation when feasible, or methods designed for off-policy stability, such as gradient-TD or emphatic TD, under their own assumptions.

Sutton & Barto (2018), Sections 11.3, 11.7–11.8.

Stability of conventional TD for policy evaluation



Reading the diagram: Stable under the corresponding convergence assumptions, including suitable step sizes. "Can diverge" means counterexamples exist, not that divergence is inevitable. The chart concerns conventional TD, not gradient-TD or emphatic methods.

Adapted from Maei (2011), *Gradient Temporal-Difference Learning Algorithms*, Figure 1.1, p. 6.

Table of Contents

Stochastic systems and value equations

Least-squares minimization of TD residuals

TD(0) – learn from one transition

Eligibility traces and TD(λ) – propagating information backward

Off-policy learning – exploration and stability

Learning a control policy – policy iteration, SARSA and Q-learning

Learning a parametrized policy – policy gradients and actor–critic

☕ Going further – modern reinforcement learning algorithms

From policy evaluation to control

Note: We start with discounted-cost control. Average-cost control is discussed at the end of this section.

So far, we have learned to evaluate a fixed policy from observed transitions and costs. We now want to use these estimates to learn a policy with lower cost.

The policy-improvement principle suggests learning action values and choosing actions with small estimated cost:

$$\phi^+(x) \in \arg\min_{u \in \mathcal{U}} Q^\theta(x, u).$$

Learning also requires exploration: the data must contain enough information about alternative actions.

We first separate policy evaluation and improvement, then interleave learning and action selection through SARSA and Q-learning.

Approximate policy iteration

Algorithm

Start from a stationary policy ϕ_0. At outer iteration j:

1. **Evaluate:** Fit $Q^{\theta_j} \approx Q_{\phi_j}$ using TD or LSTD with suitable data and sampling assumptions
2. **Improve:** Choose a measurable greedy policy

$$\phi_{j+1}(x) \in \arg\min_{u \in \mathcal{U}} Q^{\theta_j}(x, u).$$

3. Collect further data and repeat

Exact evaluation gives policy improvement under the earlier assumptions. Approximation errors can destroy monotone improvement and cause cycling. Evaluation must target ϕ_j, even if exploratory data come from a different behavior policy.

Approximate policy iteration can cycle (1/2)

Keep the two-state dynamics: from either state, r goes to state 1 and a goes to state 2. Set $\gamma = 0.9$, $c(1, u) = 1$, $c(2, u) = 2$, and use

$$h^\theta(1) = \theta, \qquad h^\theta(2) = -\theta.$$

Consider the policies $\phi_r(x) = r$ and $\phi_a(x) = a$.

Approximate evaluation: Solve the projected Bellman equation exactly in each policy's invariant distribution, as in on-policy linear TD.

Policy	Recurrent state	Projected evaluation
ϕ_r	1	$\theta = 1 + 0.9\theta \;\Rightarrow\; \theta_r = 10$
ϕ_a	2	$-\theta = 2 - 0.9\theta \;\Rightarrow\; \theta_a = -20$

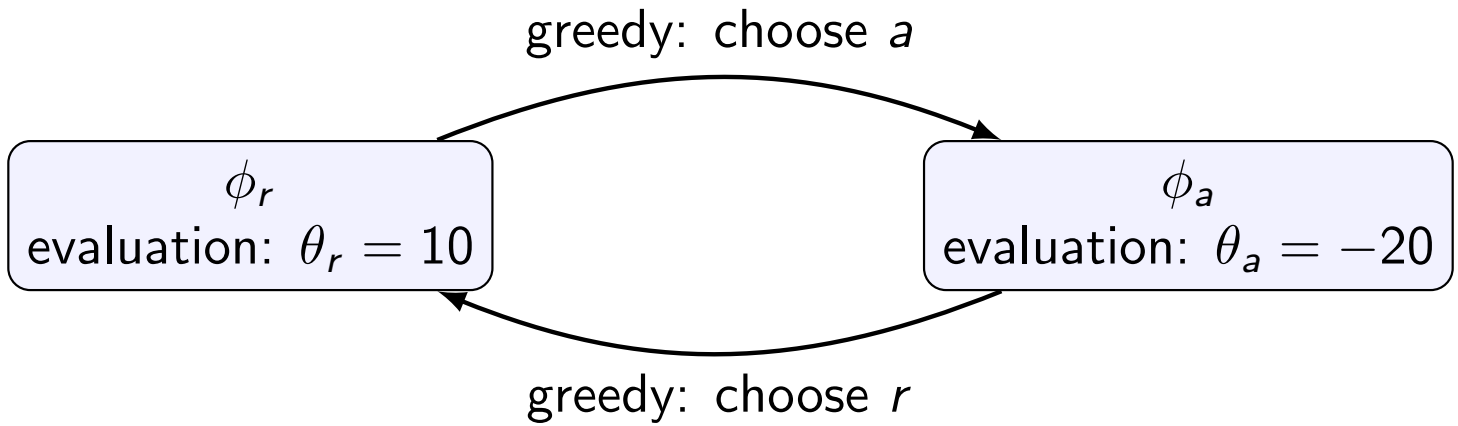
Each evaluation problem has a unique, stable TD fixed point. The approximation is accurate at the recurrent state but extrapolates poorly to the other state.

Approximate policy iteration can cycle (2/2)

Using the known transitions, reconstruct action-value estimates from h^θ:

$$\widehat{Q}_\theta(x, r) = c(x) + 0.9\theta, \qquad \widehat{Q}_\theta(x, a) = c(x) - 0.9\theta.$$

Greedy improvement selects a when $\theta > 0$, and r when $\theta < 0$. Thus evaluation and improvement repeat the cycle



The true values are

$$(h_{\phi_r}(1), h_{\phi_r}(2)) = (10, 11), \qquad (h_{\phi_a}(1), h_{\phi_a}(2)) = (19, 20).$$

From policy iteration to online control

Instead of completing policy evaluation before improvement, we now interleave learning and action selection. The action-selection policy changes as the value estimate is updated.

Assume finitely many actions and choose a greedy action

$$g_k(x) \in \arg\min_{u \in \mathcal{U}} Q^{\theta_k}(x, u).$$

An ϵ_k-greedy policy uses

$$\phi_k(u \mid x) = (1 - \epsilon_k)\mathbf{1}_{\{u = g_k(x)\}} + \frac{\epsilon_k}{|\mathcal{U}|}.$$

GLIE: Greedy in the limit with infinite exploration means that every state–action pair is visited infinitely often, while the probability of choosing a nongreedy action tends to zero. Merely choosing $\epsilon_k \to 0$ does not ensure infinite exploration.

A sufficient exploration schedule for GLIE

Assume every state is reachable from every other state under suitable actions. Let $N_k(x)$ count visits to state x, including the current visit:

$$N_k(x) = \sum_{j=0}^{k} \mathbf{1}_{\{X_j = x\}}.$$

At the current state X_k, use uniform random exploration with probability

$$\epsilon_k = \frac{\beta}{N_k(X_k)}, \qquad 0 < \beta < 1.$$

On the n-th visit to a state, each action has probability at least $\beta/(|\mathcal{U}|n)$. These probabilities sum to infinity, while exploration tends to zero as the visit count grows. Under the stated assumption, this schedule is GLIE.

A global schedule $\epsilon_k = 1/k$ is not sufficient in general.

Singh, Jaakkola, Littman & Szepesvári (2000), Appendix B.2.

SARSA(0) – update the policy after each transition

Algorithm

Initialize θ_0, observe X_0, and choose U_0. At step k, observe transition (X_k, c_k, X_{k+1}) and choose U_{k+1} using the current policy, e.g., ϵ_k-greedy with respect to Q^{θ_k}. Update

$$\begin{aligned} \delta_k &= c_k + \gamma Q^{\theta_k}(X_{k+1}, U_{k+1}) - Q^{\theta_k}(X_k, U_k), \\ \theta_{k+1} &= \theta_k + \alpha_k \delta_k \psi(X_k, U_k). \end{aligned}$$

Execute the already selected action U_{k+1} at the next step.

On-policy: The next action in the bootstrap target is the one executed, including exploration. The name refers to the tuple $(X_k, U_k, c_k, X_{k+1}, U_{k+1})$.

Q-learning – learn optimal action values while exploring

Algorithm

Initialize θ_0 and choose an exploratory behavior policy. At step k, observe transition (X_k, c_k, X_{k+1}) and update

$$\delta_k = c_k + \gamma \min_{a \in \mathcal{U}} Q^{\theta_k}(X_{k+1}, a) - Q^{\theta_k}(X_k, U_k),$$
$$\theta_{k+1} = \theta_k + \alpha_k \delta_k \psi(X_k, U_k).$$

Choose the next action using behavior, which may remain exploratory.

Off-policy control: The bootstrap target uses the smallest estimated Q-value at the next state, regardless of the action actually chosen by the behavior policy. Its intended approximation target is the Q-optimality equation

$$Q_\star(x, u) = c(x, u) + \gamma (P_u m_{Q_\star})(x).$$

SARSA versus Q-learning

Both methods use $\theta_{k+1} = \theta_k + \alpha_k \delta_k \psi(X_k, U_k)$. Their difference is the next-state prediction in δ_k.

	SARSA(0)	**Q-learning**
Prediction	$Q^{\theta_k}(X_{k+1}, U_{k+1})$	$\min_a Q^{\theta_k}(X_{k+1}, a)$
Target policy	Current exploratory policy	Current greedy policy
Sampling	On-policy	Off-policy
Exploration	In the target policy	In the behavior policy

For a tabular representation, the parameter update becomes

$$Q_{k+1}(X_k, U_k) = Q_k(X_k, U_k) + \alpha_k \delta_k,$$

with all other entries unchanged.

SARSA and Q-learning – tabular convergence

A sufficient setting is a finite state and action space, $0 < \gamma < 1$, and infinite visits to every state–action pair. Use $0 < \alpha_k \leq 1$ with, almost surely, for every $(x, u) \in \mathcal{X} \times \mathcal{U}$,

$$\sum_{k:\,(X_k,U_k)=(x,u)} \alpha_k = \infty, \qquad \sum_{k:\,(X_k,U_k)=(x,u)} \alpha_k^2 < \infty.$$

Theorem

Under these conditions: tabular Q-learning satisfies $Q_k \to Q_\star$ almost surely; tabular SARSA(0) also satisfies $Q_k \to Q_\star$ almost surely if its action-selection policy is GLIE.

Watkins & Dayan (1992); Singh et al. (2000), Theorem 1.

Tabular convergence: Bellman contraction

Define the Bellman optimality operator

$$(\mathcal{T}_\star Q)(x,u) = c(x,u) + \gamma (P_u m_Q)(x), \qquad m_Q(x) = \min_a Q(x,a).$$

For any two action-value functions,

$$|m_Q(x) - m_{\widetilde{Q}}(x)| \le \max_a |Q(x,a) - \widetilde{Q}(x,a)| \le \|Q - \widetilde{Q}\|_\infty.$$

Taking the transition expectation gives

$$\|\mathcal{T}_\star Q - \mathcal{T}_\star \widetilde{Q}\|_\infty \le \gamma \|Q - \widetilde{Q}\|_\infty.$$

Thus $\mathcal{T}_\star$ has a unique fixed point, $Q_\star$.

We will show that Q-learning is a sampled iteration of this contraction, and that SARSA adds a conditional-mean error that vanishes under GLIE.

Tabular convergence: boundedness

Let $|c_k| \le C$ and choose $B = \max\left\{ \|Q_0\|_\infty, \frac{C}{1-\gamma} \right\}$.

If $\|Q_k\|_\infty \le B$, either bootstrap target,

$$Y_k = c_k + \gamma \min_a Q_k(X_{k+1}, a) \quad \text{or} \quad Y_k = c_k + \gamma Q_k(X_{k+1}, U_{k+1}),$$

satisfies $|Y_k| \le C + \gamma B \le B$.

Write $i_k = (X_k, U_k)$. The selected entry is updated by

$$Q_{k+1}(i_k) = (1 - \alpha_k) Q_k(i_k) + \alpha_k Y_k, \qquad 0 < \alpha_k \le 1.$$

All other entries are unchanged, so induction gives

$$\|Q_k\|_\infty \le B, \qquad |Y_k| \le B \quad \text{for every } k.$$

Tabular convergence: stochastic-approximation lemma

Let $\mathcal{F}_k$ contain the history through the current pair i_k, before observing the next cost, state and action. The step size is chosen using this history.

Theorem (Asynchronous stochastic approximation)

Consider $q_k \in \mathbb{R}^d$, with finite initial value q_0 and uniformly bounded targets $|Y_k| \leq K$, updated only at i_k by

$$q_{k+1}(i_k) = (1 - \alpha_k) q_k(i_k) + \alpha_k Y_k.$$

Suppose the per-coordinate step-size conditions hold and, for some $0 \leq \kappa < 1$ and $\varepsilon_k \to 0$ almost surely,

$$|\mathbb{E}[Y_k \mid \mathcal{F}_k] - q_\star(i_k)| \leq \kappa \|q_k - q_\star\|_\infty + \varepsilon_k.$$

Then $q_k \to q_\star$ almost surely.

Bounded-target specialization of Singh et al. (2000), Lemma 1.

Tabular Q-learning convergence

For Q-learning, the conditional mean of the bootstrap target is

$$\mathbb{E}[Y_k \mid \mathcal{F}_k] = (\mathcal{T}_\star Q_k)(i_k).$$

Since $\mathcal{T}_\star Q_\star = Q_\star$, contraction yields

$$\begin{aligned} |\mathbb{E}[Y_k \mid \mathcal{F}_k] - Q_\star(i_k)| &= |(\mathcal{T}_\star Q_k)(i_k) - (\mathcal{T}_\star Q_\star)(i_k)| \\ &\leq \gamma \|Q_k - Q_\star\|_\infty. \end{aligned}$$

The stochastic-approximation lemma applies with $\kappa = \gamma$ and $\varepsilon_k = 0$. The bound $|Y_k| \leq B$ supplies its bounded-target assumption; the per-pair step sizes supply its update conditions. Therefore, $Q_k \longrightarrow Q_\star$ almost surely.

Tabular SARSA convergence under GLIE

SARSA's bootstrap target differs from Q-learning's by a nonnegative term:

$$Y_k = c_k + \gamma \min_a Q_k(X_{k+1}, a) + \gamma D_k,$$

$$D_k = Q_k(X_{k+1}, U_{k+1}) - \min_a Q_k(X_{k+1}, a).$$

Let p_k be the largest nongreedy-action probability over next states. For finitely many states, GLIE gives $p_k \to 0$ almost surely. Since $D_k = 0$ for a greedy action and $0 \le D_k \le 2B$,

$$0 \le b_k := \gamma \mathbb{E}[D_k \mid \mathcal{F}_k] \le 2\gamma B p_k \longrightarrow 0.$$

Consequently,

$$|\mathbb{E}[Y_k \mid \mathcal{F}_k] - Q_\star(i_k)| \le \gamma \|Q_k - Q_\star\|_\infty + b_k.$$

The same SA lemma, with $\varepsilon_k = b_k$, yields $Q_k \to Q_\star$ almost surely.

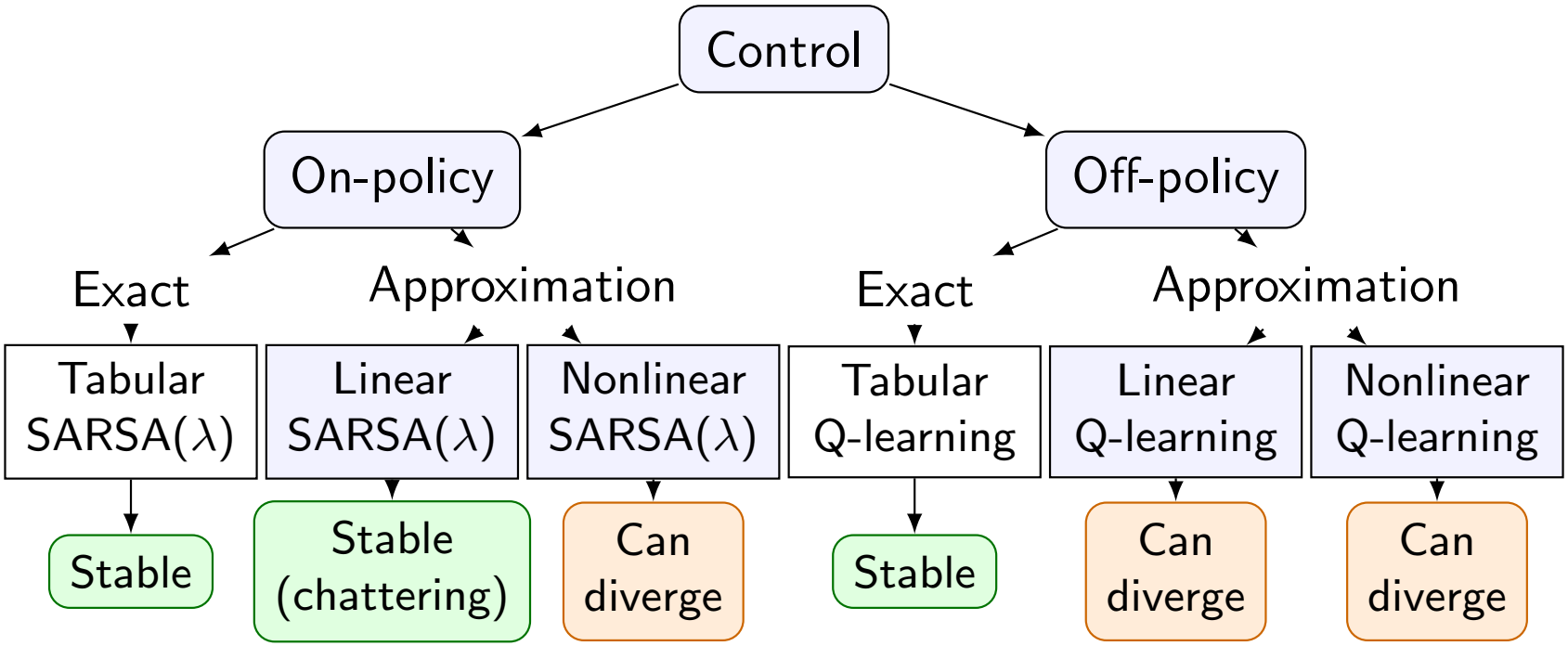
Control with function approximation – what carries over?

The tabular convergence guarantees do not transfer automatically.

- **Policy iteration:** Errors in the estimated Q-values can make a greedy policy change increase the actual cost
- **SARSA:** The fixed-policy TD theorem does not apply directly to control. Under suitable assumptions, linear SARSA remains bounded but may chatter rather than converge to a fixed solution
- **Q-learning:** Off-policy sampling, bootstrapping and shared parameters form the deadly triad
- **Traces:** Off-policy trace methods require their own algorithms and assumptions; the preceding results concern one-step control

Sutton & Barto (2018), Chapters 10–12. Gordon (2000), *Reinforcement Learning with Function Approximation Converges to a Region*. Gordon's result concerns trajectory-based SARSA(0), with the policy held fixed within each episode and allowed to change between episodes.

Stability of conventional TD for control



Reading the diagram: Stability is conditional on the relevant assumptions; it does not imply convergence to an optimal policy. Linear SARSA may chatter: estimates can remain bounded while the greedy policy keeps switching. "Can diverge" means counterexamples exist.

Adapted from Maei (2011), *Gradient Temporal-Difference Learning Algorithms*, Figure 1.1, control branch, p. 6.

A bridge to average-cost control

We now replace the discounted objective by the average cost η_ϕ. Under the regularity and verification assumptions stated earlier, the optimality equation is

$$\eta_\star + Q_\star(x,u) = c(x,u) + (P_u m_{Q_\star})(x),$$

where $m_q(x) = \min_a q(x,a)$ and $\eta_\star = \inf_\phi \eta_\phi$. A minimizing selector gives an optimal policy under those assumptions.

Average cost and relative values: Learning must identify both the scalar average cost and relative action values. Adding a constant to $Q_\star$ leaves the equation and greedy actions unchanged. Fix that freedom, e.g., by imposing $Q_\star(x_{\mathrm{ref}}, u_{\mathrm{ref}}) = 0$.

Average-cost control: algorithms

Differential SARSA: Use the next action actually chosen:

$$\delta_k = c_k - \rho_k + Q^{\theta_k}(X_{k+1}, U_{k+1}) - Q^{\theta_k}(X_k, U_k),$$

Differential Q-learning: Use the smallest next-state prediction:

$$\delta_k = c_k - \rho_k + \min_a Q^{\theta_k}(X_{k+1}, a) - Q^{\theta_k}(X_k, U_k),$$

Differential updates using the corresponding δ_k are

$$\theta_{k+1} = \theta_k + \alpha_k \delta_k \psi(X_k, U_k), \qquad \rho_{k+1} = \rho_k + \beta_k \delta_k.$$

The average-cost estimate uses the TD error, not just $c_k - \rho_k$.

Average-cost control: tabular convergence?

Setting $\gamma = 1$ loses the strict contraction used in the discounted proof. Average-cost results require additional structure.

Differential Q-learning: Choose $\beta_k = \xi\alpha_k$, with fixed $\xi > 0$. For a finite MDP where every state is reachable from every other state under suitable actions, assume the optimality equation has a unique Q-solution up to a constant. Under the per-pair step-size and update-frequency conditions,

$$\rho_k \longrightarrow \eta_\star, \qquad Q_k \longrightarrow Q_\star \quad \text{almost surely,}$$

where $Q_\star$ solves the average-cost Q-optimality equation.

Differential SARSA: The policy and average-cost estimate evolve together. Neither the discounted GLIE theorem nor fixed-policy average-cost TD theory directly proves convergence of these joint control updates.

Sutton & Barto (2018), Sections 10.3–10.5; Wan, Naik & Sutton (2021), Theorem 1 and Appendix B.

TD control – strengths and limitations

- **Online and model-free:** SARSA and Q-learning learn from individual transitions, without a transition model or complete returns
- **Different uses of exploration:** SARSA accounts for exploratory actions in its target; Q-learning learns greedy action values from exploratory behavior
- **Exploration and scale:** Learning requires sufficient state–action coverage. Tables grow with the state and action spaces; greedy action selection can be costly for large or continuous action spaces
- **Approximation can destabilize control:** Value errors can lead to poor policy changes or cycling. Combining off-policy learning, bootstrapping and function approximation can cause divergence

The discounted tabular guarantees require suitable exploration and step sizes. Function approximation and average-cost control need separate analysis.

Table of Contents

Stochastic systems and value equations

Least-squares minimization of TD residuals

TD(0) – learn from one transition

Eligibility traces and TD(λ) – propagating information backward

Off-policy learning – exploration and stability

Learning a control policy – policy iteration, SARSA and Q-learning

Learning a parametrized policy – policy gradients and actor–critic

☕ Going further – modern reinforcement learning algorithms

From greedy improvement to policy gradients

Note: We start with discounted-cost policy gradients. Average-cost policy gradients are discussed at the end of this section.

So far, policy improvement has selected actions by minimizing an estimated Q-function. We now represent the policy directly by parameters $\theta \in \mathbb{R}^p$ and improve these parameters by gradient descent.

- **Policy gradient:** Estimate how the cost changes with θ
- **REINFORCE:** Estimate this gradient using sampled returns
- **Actor–critic:** Use a learned value function to estimate the gradient with less reliance on long rollouts

This avoids an action minimization at each update and accommodates continuous actions. Optimization is over the chosen policy family; global optimality is generally not guaranteed.

Parametrized policies and their score

Let $\phi^\theta(u \mid x)$ denote a differentiable policy probability or density with parameter-independent support. Its **score** is

$$s_\theta(x, u) := \nabla_\theta \log \phi^\theta(u \mid x) \in \mathbb{R}^p.$$

Interpretation: The score $s_\theta(x, u)$ points in the parameter direction that locally increases the log-probability of action u at state x. A policy-gradient descent update weights this direction by the negative estimated advantage: actions estimated to perform better than the current policy average are encouraged; worse actions are discouraged.

Two useful properties:

- $\nabla_\theta \phi^\theta(u \mid x) = \phi^\theta(u \mid x) s_\theta(x, u)$
- $\mathbb{E}[s_\theta(X, U) \mid X = x] = \int_{\mathcal{U}} \phi^\theta(u \mid x) s_\theta(x, u)\, du = 0$

Here integration means summation for discrete actions.

Parametrized policies for finite and continuous actions

Finite actions: With policy features $\chi(x,u) \in \mathbb{R}^p$, a softmax policy:

$$\phi^\theta(u \mid x) = \frac{\exp(\chi(x,u)^\top \theta)}{\sum_{a \in \mathcal{U}} \exp(\chi(x,a)^\top \theta)},$$
$$s_\theta(x,u) = \chi(x,u) - \sum_{a \in \mathcal{U}} \phi^\theta(a \mid x)\chi(x,a).$$

Continuous actions: For $\mathcal{U} = \mathbb{R}^m$, a Gaussian policy with fixed covariance:

$$U \mid X = x \sim \mathcal{N}(m_\theta(x), \Sigma), \qquad \Sigma \succ 0.$$

Writing $D_\theta m_\theta(x) \in \mathbb{R}^{m \times p}$ for the Jacobian,

$$s_\theta(x,u) = D_\theta m_\theta(x)^\top \Sigma^{-1}(u - m_\theta(x)).$$

Discounted objective and state weighting

Fix an initial distribution μ, independent of θ, and $0 < \gamma < 1$. Write $h_\theta = h_{\phi^\theta}$ and $Q_\theta = Q_{\phi^\theta}$ for the true discounted values. Our objective is

$$J_\gamma(\theta) = \mathbb{E}_\mu^{\phi^\theta}\left[\sum_{t=0}^{\infty} \gamma^t c_t\right] = \int_{\mathcal{X}} h_\theta(x)\, \mu(dx).$$

Define the normalized discounted state distribution

$$d_{\gamma,\theta}(B) = (1-\gamma) \sum_{t=0}^{\infty} \gamma^t \underbrace{\mathbb{P}_\mu^{\phi^\theta}(X_t \in B)}_{= (\mu P_{\phi^\theta}^t)(B)}.$$

This distribution depends on the policy and on μ. It generally differs from the invariant distribution π_θ.

The discounted policy-gradient theorem

A sufficient setting for the policy-gradient theorem is:

- Bounded cost: $|c(x,u)| \le C$ for all x, u.
- $\phi^\theta(u \mid x)$ is positive on a common support independent of θ. For each (x,u) in this support, $\phi^\theta(u \mid x)$ is continuously differentiable in θ
- Bounded score: $\sup_\theta \|s_\theta(x,u)\| \le L$ for all (x,u) in the support

Theorem

Under these conditions, for $X \sim d_{\gamma,\theta}$ and $U \sim \phi^\theta(\cdot \mid X)$,

$$\boxed{\nabla J_\gamma(\theta) = \frac{1}{1-\gamma}\mathbb{E}[s_\theta(X,U)Q_\theta(X,U)].}$$

Discounted policy-gradient theorem: cost gradient

Starting from $X_0 = x$ and following ϕ^θ, let $M_{\theta,x}$ be the law of $\tau_t = (x_0, u_0, \dots, x_t, u_t)$. Define the expected cost at time t by

$$m_{t,\theta}(x) := \mathbb{E}^{\phi^\theta}[c(X_t, U_t) \mid X_0 = x] = \int_{\mathcal{D}} c(x_t, u_t)\, M_{\theta,x}(d\tau_t),$$

where $\mathcal{D} = (\mathcal{X} \times \mathcal{U})^{t+1}$. Only the policy factors $\prod_{j=0}^{t} \phi^\theta(u_j \mid x_j)$ depend on θ, so

$$\nabla_\theta m_{t,\theta}(x) = \int_{\mathcal{D}} c(x_t, u_t) \left(\sum_{j=0}^{t} s_\theta(x_j, u_j) \right) M_{\theta,x}(d\tau_t).$$

The triangle inequality and the cost and score bounds imply

$$\|\nabla_\theta m_{t,\theta}(x)\| \le \int C \sum_{j=0}^{t} L\, M_{\theta,x}(d\tau_t) = CL(t+1).$$

Discounted policy-gradient theorem: domination

Since $h_\theta(x) = \sum_{t=0}^{\infty} \gamma^t m_{t,\theta}(x)$, the preceding bound permits term-by-term differentiation. It follows that h_θ is continuously differentiable and

$$\|h_\theta\|_\infty \le \frac{C}{1-\gamma}, \qquad \|\nabla_\theta h_\theta\|_\infty \le \frac{CL}{(1-\gamma)^2}.$$

These bounds also justify $\nabla_\theta(P_u h_\theta) = P_u(\nabla_\theta h_\theta)$.

Integrating the bounded score along the parameter segment gives, for sufficiently small $r > 0$,

$$\phi^\theta(u \mid x) \le e^{Lr}\phi^{\theta_0}(u \mid x), \qquad \|\theta - \theta_0\| \le r,$$

For the Bellman integrand $B_\theta(x,u) = \phi^\theta(u \mid x)\{c(x,u) + \gamma(P_u h_\theta)(x)\}$, the product rule yields

$$\|\nabla_\theta B_\theta(x,u)\| \le \frac{CL}{(1-\gamma)^2} e^{Lr}\phi^{\theta_0}(u \mid x).$$

Discounted policy-gradient theorem: proof (1/2)

In the Bellman equation,

$$h_\theta(x) = \int_{\mathcal{U}} \phi^\theta(u \mid x) \underbrace{\{c(x,u) + \gamma(P_u h_\theta)(x)\}}_{Q_\theta(x,u)} \, du,$$

the above domination permits differentiation inside the integral:

$$\nabla_\theta h_\theta(x) = \underbrace{\int_{\mathcal{U}} \phi^\theta(u \mid x) s_\theta(x,u) Q_\theta(x,u) \, du}_{g_\theta(x)} + \gamma \int_{\mathcal{U}} \phi^\theta(u \mid x)(P_u \nabla_\theta h_\theta)(x) \, du.$$

Thus $\nabla_\theta h_\theta = g_\theta + \gamma P_{\phi^\theta} \nabla_\theta h_\theta$. The two terms describe changing the current action probabilities and changing the policy at future states, respectively.

Discounted policy-gradient theorem: proof (2/2)

A Markov operator satisfies $\|P_{\phi^\theta}\|_\infty \leq 1$. Thus

$$(I - \gamma P_{\phi^\theta})^{-1} = \sum_{t=0}^{\infty} \gamma^t P_{\phi^\theta}^t.$$

Applying this operator on the previous equation gives

$$\nabla_\theta h_\theta = \sum_{t=0}^{\infty} \gamma^t P_{\phi^\theta}^t g_\theta.$$

Finally, integrating against μ gives

$$\nabla J_\gamma(\theta) = \sum_{t=0}^{\infty} \gamma^t \mathbb{E}_\mu^{\phi^\theta}[g_\theta(X_t)] = \frac{1}{1-\gamma} \mathbb{E}_{d_{\gamma,\theta},\phi^\theta}[s_\theta Q_\theta].$$

Note: No invariant distribution or ergodicity assumption is needed.

Baselines and the advantage function

For any integrable state-dependent baseline b,

$$\mathbb{E}[s_\theta(X, U)b(X)] = 0.$$

Consequently,

$$\nabla J_\gamma(\theta) = \frac{1}{1-\gamma}\mathbb{E}[s_\theta(X, U)\{Q_\theta(X, U) - b(X)\}].$$

Purpose: A suitable baseline can reduce variance without changing the mean gradient.

Taking $b = h_\theta$ defines the **advantage**

$$A_\theta(x, u) := Q_\theta(x, u) - h_\theta(x).$$

A negative advantage means that action u has lower expected cost than the current policy's average action at x.

REINFORCE – learn from complete returns

Consider episodes ending in an **absorbing zero-cost state** in finite time almost surely. Start at $X_0 \sim \mu$ and freeze θ_j during episode j.

Algorithm

Generate an episode under ϕ^{θ_j} and compute the complete returns

$$G_t = \sum_{\ell=t}^{T-1} \gamma^{\ell-t} c_\ell, \qquad 0 \le t < T.$$

With a baseline b_j fixed before the episode, update

$$\theta_{j+1} = \theta_j - \alpha_j \sum_{t=0}^{T-1} \gamma^t s_{\theta_j}(X_t, U_t)\{G_t - b_j(X_t)\}.$$

Complete returns avoid bootstrapping but can have high variance. Truncating continuing tasks generally introduces bias.

From complete returns to a TD critic

Freeze the policy parameter θ and approximate its value by

$$h^{\omega}(x) = \psi(x)^{\top}\omega, \qquad \omega \in \mathbb{R}^{d}.$$

At step k, observe transition (X_k, U_k, c_k, X_{k+1}) under ϕ^{θ} and define

$$\delta_k = c_k + \gamma h^{\omega}(X_{k+1}) - h^{\omega}(X_k).$$

The critic's **bootstrap target** is $c_k + \gamma h^{\omega}(X_{k+1})$.

Proposition

If $h^{\omega} = h_{\theta}$, then

$$\mathbb{E}[\delta_k \mid X_k = x, U_k = u] = A_{\theta}(x, u).$$

Indeed, the conditional expectation of the first two terms is $Q_{\theta}(x, u)$. The actor can therefore replace a complete return by a TD error when the critic is exact.

Discounted actor–critic – sampling the right states

The gradient formula weights states by $d_{\gamma,\theta}$. One way to obtain this weighting is to restart the state process independently with probability $1-\gamma$ after each transition.

Sampling rule: Draw $U_k \sim \phi^{\theta_k}(\cdot \mid X_k)$ and observe the physical transition (X_k, U_k, c_k, X'_k). After the learning update, choose

$$X_{k+1} = \begin{cases} X'_k, & \text{with probability } \gamma, \\ \widetilde{X}_k, \quad \widetilde{X}_k \sim \mu, & \text{with probability } 1-\gamma. \end{cases}$$

For a frozen policy, the restarted chain has invariant distribution $d_{\gamma,\theta}$ because

$$d_{\gamma,\theta} = (1-\gamma)\mu + \gamma d_{\gamma,\theta} P_{\phi^\theta}.$$

Use the physical successor X'_k in the bootstrap target. This scheme requires access to resets; episodic discounted weighting is an alternative.

Discounted actor–critic – update both approximations

The **actor** is the policy ϕ^θ; the **critic** is h^ω. Use the restarted sampling scheme from the preceding slide.

Algorithm

Initialize θ_0, ω_0 and $X_0 \sim \mu$. At step k, observe transition (X_k, U_k, c_k, X_k') and update

$$\begin{aligned} \delta_k &= c_k + \gamma h^{\omega_k}(X_k') - h^{\omega_k}(X_k), \\ \omega_{k+1} &= \omega_k + \beta_k \delta_k \psi(X_k), \\ \theta_{k+1} &= \theta_k - \alpha_k s_{\theta_k}(X_k, U_k)\delta_k. \end{aligned}$$

Then restart or continue according to the sampling rule.

Two time scales: Typically the actor changes more slowly: $\alpha_k/\beta_k \to 0$. Exact-critic identities alone do not prove convergence of these joint updates.

An approximate critic can bias the actor

Freeze θ, ω, let $e = h^\omega - h_\theta$, and sample $X \sim d_{\gamma,\theta}$, $U \sim \phi^\theta$, with physical successor X'. Then

$$\mathbb{E}[\delta \mid X = x, U = u] = A_\theta(x, u) + \gamma(P_u e)(x) - e(x).$$

The state baseline $e(X)$ cancels, but the next-state error need not:

$$\frac{1}{1-\gamma}\mathbb{E}[s_\theta(X, U)\delta] = \nabla J_\gamma(\theta) + \frac{\gamma}{1-\gamma}\mathbb{E}[s_\theta(X, U)e(X')].$$

Consequence: Convergence of the critic to a projected Bellman solution does not by itself make the actor update an unbiased policy gradient. A tabular exact critic, suitable compatible critics, or bounds on critic error provide different ways to analyze the actor.

Average-cost policy gradients

Now let $J_{\mathrm{av}}(\theta) = \eta_{\phi^\theta}$ and let π_θ be the invariant state distribution. Write h_θ, Q_θ for the *relative* values from this slide onward:

$$Q_\theta(x,u) = c(x,u) - \eta_{\phi^\theta} + (P_u h_\theta)(x).$$

The relative value solves the Poisson equation

$$\eta_{\phi^\theta} + h_\theta = c_{\phi^\theta} + P_{\phi^\theta} h_\theta .$$

Fix its additive constant, for example by $h_\theta(x_{\mathrm{ref}}) = 0$.

We retain the bounded-cost and smooth-policy assumptions from the discounted theorem, allowing continuous states and actions. Existence and regularity of the relative value require additional assumptions; they no longer follow from discounting.

The average-cost policy-gradient theorem

Reuse the bounded-cost, common-support, differentiability and bounded-score assumptions of the discounted theorem. Also assume:

- Each policy has a unique invariant distribution π_θ
- The normalized Poisson solution $h_\theta(x)$ exists and is continuously differentiable in θ for every $x \in \mathcal{X}$, with

$$\sup_\theta \|h_\theta\|_\infty \le H, \qquad \sup_\theta \|\nabla_\theta h_\theta\|_\infty \le D.$$

Theorem

Under these conditions, for $X \sim \pi_\theta$ and $U \sim \phi^\theta(\cdot \mid X)$,

$$\boxed{\nabla J_{\mathrm{av}}(\theta) = \mathbb{E}[s_\theta(X,U) Q_\theta(X,U)] = \mathbb{E}[s_\theta(X,U) A_\theta(X,U)].}$$

Here $A_\theta = Q_\theta - h_\theta$, as in the discounted case.

Average-cost policy-gradient theorem: proof (1/2)

Reuse the discounted domination argument with H, D in place of the discounted value bounds. For $B_\theta(x,u) = \phi^\theta(u \mid x)\{c(x,u) + (P_u h_\theta)(x)\}$,

$$\|\nabla_\theta B_\theta(x,u)\| \le \{L(C+H)+D\} e^{Lr} \phi^{\theta_0}(u \mid x)$$

when $\|\theta - \theta_0\| \le r$ is sufficiently small. This integrable bound permits differentiating the Poisson equation:

$$\eta_{\phi^\theta} + h_\theta(x) = \int_{\mathcal{U}} \phi^\theta(u \mid x)\{c(x,u) + (P_u h_\theta)(x)\}\, du.$$

As in the discounted proof, use $\nabla_\theta \phi^\theta = \phi^\theta s_\theta$ and $\nabla_\theta P_u h_\theta = P_u \nabla_\theta h_\theta$. The result is

$$\nabla_\theta \eta_{\phi^\theta} + \nabla_\theta h_\theta = g_\theta + P_{\phi^\theta} \nabla_\theta h_\theta,$$

where $g_\theta(x) = \int_{\mathcal{U}} \phi^\theta(u \mid x) s_\theta(x,u) Q_\theta(x,u)\, du$.

Average-cost policy-gradient theorem: proof (2/2)

Integrate the differentiated equation against π_θ. Boundedness of $\nabla_\theta h_\theta$ and invariance give

$$\pi_\theta(P_{\phi^\theta} \nabla_\theta h_\theta) = \pi_\theta(\nabla_\theta h_\theta).$$

The relative-value derivative terms cancel, leaving

$$\nabla J_{\mathrm{av}}(\theta) = \pi_\theta(g_\theta) = \mathbb{E}_{\pi_\theta, \phi^\theta}[s_\theta(X, U) Q_\theta(X, U)].$$

The baseline identity from the discounted case replaces Q_θ by $A_\theta = Q_\theta - h_\theta$ without changing the expectation.

Difference from discounted cost: Invariance replaces the discounted resolvent; setting $\gamma = 1$ in that resolvent would make it singular.

Average-cost actor–critic

Use $h^{\omega}(x) = \psi(x)^{\top}\omega$ for the relative-value critic and ρ for the average cost. Fix the critic's additive constant, for example by excluding its constant feature direction.

Algorithm

Initialize $\theta_0, \omega_0, \rho_0$. At step k, sample $U_k \sim \phi^{\theta_k}(\cdot \mid X_k)$, observe transition (X_k, U_k, c_k, X_{k+1}), and update

$$\begin{aligned}
\delta_k &= c_k - \rho_k + h^{\omega_k}(X_{k+1}) - h^{\omega_k}(X_k), \\
\rho_{k+1} &= \rho_k + \beta_k(c_k - \rho_k), \\
\omega_{k+1} &= \omega_k + \beta_k \delta_k \psi(X_k), \\
\theta_{k+1} &= \theta_k - \alpha_k s_{\theta_k}(X_k, U_k)\delta_k.
\end{aligned}$$

For a frozen actor and exact critic, the stationary mean actor direction is $-\nabla J_{\mathrm{av}}(\theta)$. Unlike discounted cost, the relevant state weighting is the invariant distribution of the continuing chain.

Compatible approximation – preserve the gradient

The preceding algorithms use a state-value critic h^ω and send its TD error to the actor. Compatibility suggests an alternative: learn a **Q-critic** and send its prediction to the actor.

Let ν_θ denote the relevant state–action distribution: $d_{\gamma,\theta}\phi^\theta$ or $\pi_\theta\phi^\theta$. For a frozen actor θ, choose score features and fit

$$Q^\omega(x,u) = s_\theta(x,u)^\top \omega, \qquad \omega_\star \in \arg\min_v \mathbb{E}_{\nu_\theta}[(Q_\theta - s_\theta^\top v)^2].$$

Its least-squares normal equation gives

$$\mathbb{E}_{\nu_\theta}[s_\theta(Q_\theta - Q^{\omega_\star})] = 0, \qquad \mathbb{E}_{\nu_\theta}[s_\theta Q^{\omega_\star}] = \mathbb{E}_{\nu_\theta}[s_\theta Q_\theta].$$

Meaning: The critic preserves the gradient-relevant expectation even if it cannot represent the full Q-function.

Sutton et al. (1999), compatible function approximation.

Using a compatible Q-critic in the actor

Actor update: With the same state sampling as before, replace the TD-error signal by the Q-critic prediction:

$$\theta_{k+1} = \theta_k - \alpha_k s_{\theta_k}(X_k, U_k) Q^{\omega_k}(X_k, U_k).$$

At the fitted critic $\omega_\star$ for a frozen actor,

$$\mathbb{E}_{\nu_\theta}[s_\theta Q^{\omega_\star}] = \begin{cases} (1-\gamma)\nabla J_\gamma(\theta), & \text{discounted cost,} \\ \nabla J_{\mathrm{av}}(\theta), & \text{average cost.} \end{cases}$$

Critic fitting: The critic must track the least-squares fit from the preceding slide as θ changes. Regression on unbiased Q estimates gives the required normal equation. On-policy Q-TD(1) can also provide this fit when its approximation target is the projection under ν_θ, as detailed next.

Q-TD(1) and compatible approximation

Freeze the actor θ and use Q-critic features whose span contains every component of s_θ. If the Q-TD(1) approximation target is the least-squares projection under ν_θ, then

$$Q^{\omega_\star} = \Pi_{\nu_\theta} Q_\theta \quad \Longrightarrow \quad \mathbb{E}_{\nu_\theta}[s_\theta (Q_\theta - Q^{\omega_\star})] = 0.$$

This is exactly the orthogonality required by the actor update.

- **Discounted cost:** The projection must use $d_{\gamma,\theta}\phi^\theta$. Ordinary continuing-trajectory TD(1) generally uses the invariant state–action distribution instead; discounted sampling and trace handling must be justified together
- **Average cost:** Undamped traces require additional recurrence and trace assumptions. Konda–Tsitsiklis use a Q-critic with traces reset at a reference state and features containing the score components

Konda & Tsitsiklis (2003), Sections 4.2.1 and 5.1, Lemma 6.1.

Actor–critic – what convergence requires

Two time scales: Typical step-size conditions are

$$\sum_k \alpha_k = \sum_k \beta_k = \infty, \qquad \sum_k (\alpha_k^2 + \beta_k^2) < \infty, \qquad \frac{\alpha_k}{\beta_k} \to 0.$$

For example, $\beta_k = (k+1)^{-b}$ and $\alpha_k = (k+1)^{-a}$ with $1/2 < b < a \leq 1$.

A convergence theorem also needs smooth policies, suitable uniform mixing and moment bounds, stable critic tracking, bounded iterates, and control of the actor's gradient bias. With an exact gradient direction, the limiting actor dynamics are

$$\dot{\theta} = -\nabla J(\theta), \qquad \frac{d}{dt} J(\theta(t)) = -\|\nabla J(\theta(t))\|^2.$$

A persistent critic bias changes these dynamics; the displayed algorithms with arbitrary features have no general convergence guarantee.

Konda & Tsitsiklis (2003), two-time-scale actor–critic theory and its assumptions.

Policy gradients – strengths and limitations

- **Direct policy optimization:** Learn a stochastic policy without an action minimization at each update; suitable parametrizations accommodate continuous actions
- **Variance and data:** Return-based gradients can have high variance. Baselines can help, but exploration and sufficient data remain essential
- **Critic tradeoff:** Actor–critic enables incremental updates and can reduce variance, but an approximate critic can bias the actor direction
- **Restricted optimization:** Performance depends on the policy family and initialization; convergence, when established, generally does not imply global optimality

A policy-gradient identity alone does not guarantee convergence: sampling, step sizes and critic tracking must also be justified.

Table of Contents

Stochastic systems and value equations

Least-squares minimization of TD residuals

TD(0) – learn from one transition

Eligibility traces and TD(λ) – propagating information backward

Off-policy learning – exploration and stability

Learning a control policy – policy iteration, SARSA and Q-learning

Learning a parametrized policy – policy gradients and actor–critic

Going further – modern reinforcement learning algorithms

From the course to deep reinforcement learning

Deep RL combines value estimation and policy improvement with neural-network representations.

- **DQN:** Neural Q-learning with stored transitions reused for training and a slowly updated network for computing TD targets.
- **PPO:** Policy gradients with controlled policy updates
- **SAC:** Off-policy actor–critic with an entropy-regularized objective, especially for continuous control
- **GRPO:** A policy-optimization variant using groups of sampled responses, illustrated through language-model training

Reward convention for this section: Maximize rewards $r_k = -c_k$. Reuse h, Q and $A = Q - h$ for reward values: their signs are opposite to the earlier cost values for the same policy and discount.

As before, θ parametrizes the actor and ω the critic. Positive advantage now means *better* than the policy average.

DQN – making neural Q-learning more stable

Use a deep Q-network $Q^{\omega}(x, u)$ and target parameters $\bar{\omega}$. Sample a minibatch B of transitions (x_i, u_i, r_i, x_i') from a replay buffer $\mathcal{D}$.

$$y_i := r_i + \gamma \max_{a \in \mathcal{U}} Q^{\bar{\omega}}(x_i', a), \qquad \widehat{L}_B(\omega) := \frac{1}{2|B|} \sum_{i \in B} \big(Q^{\omega}(x_i, u_i) - y_i\big)^2.$$

Minimize $\widehat{L}_B$ in ω with targets y_i held fixed; periodically copy $\bar{\omega} \leftarrow \omega$.

- **Replay:** Reuse past transitions and randomize training batches to reduce correlations between consecutive observations
- **Exploration:** Keep collecting new transitions, e.g., with an ϵ-greedy behavior policy

These changes help in practice but do not remove the deadly triad. Maximizing over actions also limits to manageable action sets.

Mnih et al. (2015), Human-level control through deep reinforcement learning.

GAE – the connection back to TD(λ)

On a rollout under a fixed policy, freeze the critic h^ω and compute

$$\delta_k := r_k + \gamma h^\omega(X_{k+1}) - h^\omega(X_k).$$

For a rollout segment ending at T, generalized advantage estimation uses

$$\widehat{A}_k := \sum_{\ell=0}^{T-k-1} (\gamma\lambda)^\ell \delta_{k+\ell}, \qquad 0 \le \lambda \le 1.$$

Idea: Combine future prediction errors to assess the action at time k, including delayed consequences; later errors receive less weight

- $\lambda = 0$: one TD error—typically lower variance, but stronger reliance on the critic's accuracy
- $\lambda = 1$: the sum telescopes into the bootstrapped return minus $h^\omega(X_k)$—less reliance on the critic's accuracy, but typically higher variance

Schulman et al. (2016), generalized advantage estimation.

PPO – limit the incentive for large policy changes

Collect data under $\phi^{\theta_{\text{old}}}$ and freeze the advantage estimates. Define the action-probability ratio and clipped surrogate

$$w_k(\theta) := \frac{\phi^{\theta}(U_k \mid X_k)}{\phi^{\theta_{\text{old}}}(U_k \mid X_k)},$$

$$\widehat{L}_{\text{clip}}(\theta) := \frac{1}{|B|} \sum_{k \in B} \min\Big\{ w_k(\theta)\widehat{A}_k,\ \text{clip}(w_k(\theta), 1-\epsilon, 1+\epsilon)\widehat{A}_k \Big\}.$$

Here B indexes the rollout batch and $0 < \epsilon < 1$ is the clipping threshold.

1. Estimate advantages, commonly with GAE
2. Maximize $\widehat{L}_{\text{clip}}$ over several minibatch passes; also fit the critic
3. Refresh the behavior policy and collect a new batch

Clipping removes incentives for some excessive changes, especially in the improvement direction.

Schulman et al. (2017), Proximal policy optimization algorithms.

Language-model fine-tuning as reinforcement learning

A pretrained language model defines a policy over the next token. For a prompt $q \sim \mu$ and response $o = (o_0, \ldots, o_{T-1})$:

RL object	Language-model example
State X_k	Prompt and response prefix $(q, o_{<k})$
Action U_k	Next token o_k
Transition	Append the selected token
Episode	Generate until termination or a length limit
Reward	Score the response with a verifier or reward model

With a terminal score $R(q, o)$, define the response law and objective:

$$\Phi^\theta(o \mid q) := \prod_{k=0}^{T-1} \phi^\theta(o_k \mid q, o_{<k}), \qquad \max_\theta \ \mathbb{E}_{\mu,\, \Phi^\theta}[R(q, o)].$$

This is a finite-response episodic objective, not average-cost control. A reference-policy penalty can additionally discourage excessive drift.

GRPO – compare responses to the same prompt

In group relative policy optimization, sample $G > 1$ responses o_i to the same prompt q under $\Phi^{\theta_{\text{old}}}$ and score them $R_i = R(q, o_i)$. Form group-relative signals

$$\overline{R} := \frac{1}{G} \sum_{i=1}^{G} R_i, \qquad \widehat{A}_i := \frac{R_i - \overline{R}}{s_R + \varepsilon_{\text{num}}},$$

where s_R is the group's reward standard deviation and $\varepsilon_{\text{num}} > 0$.

Use $\widehat{A}_i$ for the tokens of response i in a PPO-style clipped update. The original formulation also penalizes deviation from a reference policy.

- **Benefit:** No separately trained value critic
- **Cost:** Multiple responses must be generated for each prompt
- **Limit:** Equal group rewards give no relative learning signal

The normalized signal is not generally an unbiased advantage estimate.

Shao et al. (2024), DeepSeekMath, Sections 4.

Modern RL – main ideas and tradeoffs

Method	Main idea	Main limitation
DQN	Reuse transitions; compute bootstrap targets with a separate network	Action maximization must be tractable; instability can remain
PPO	Estimate advantages with GAE; reuse each rollout batch for clipped policy updates	Needs fresh rollouts; clipping does not guarantee improvement
GRPO	Compare responses to the same prompt; no learned value critic	Multiple generations; equal scores give no relative signal

These examples connect the course's value estimation and policy improvement methods to deep RL. Further directions include off-policy actor–critic methods that reuse past transitions (SAC, TD3), planning with learned dynamics models, and offline RL from fixed datasets.